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SCHOOL OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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BANK NOTE AUTHENTICATION SYSTEM USING MACHINE LEARNING

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DECLARATION

I hereby declare that the project work entitled "Bank note Authentication System" submitted to the department of Information Technology in School of Computing and Information Technology is a record of original work done by me under the guidance of Ms. Judy Gateri.

The information and data given in the report is authentic to the best of my knowledge and the project proposal has not been submitted in any other institution for academic award.

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SCT221-004-0401/2020

This project proposal has been submitted for examination with my approval as the university supervisor.

Supervisor's Name: Ms. Judy Gateri

Signature: ……

Date: …….

DEDICATION

I dedicate this project report to the unwavering support and boundless love of my mother and siblings. Throughout the journey of this project, their encouragement, understanding, and sacrifices have been my pillars of strength.

To my dearest mother, your constant belief in my abilities and your selfless devotion have been the driving force behind my determination. Your sacrifices and tireless efforts to provide me with the resources I needed for this project will forever be etched in my heart.

To my siblings, who stood by me through every challenge and cheered me on during moments of triumph, your presence has been a source of inspiration. Your shared laughter, encouragement, and late-night brainstorming sessions have made this endeavor all the more meaningful.

This project is a testament to the collective strength of our family bond. As I present this report, I do so with deep gratitude for the invaluable role each of you played in its fruition. Thank you for being my constant support system, for believing in me when I doubted myself, and for being the source of my motivation.

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ABSTRACT

Counterfeit banknotes continue to pose a significant threat to global financial systems, despite the increasing prevalence of digital transactions. The ease with which counterfeiters can produce fake banknotes that closely resemble genuine currency demands an urgent solution for reliable banknote authentication. This project addresses this challenge by leveraging classification algorithms to enhance the accuracy of banknote authentication.

The primary objective is to develop a robust system using image data extracted from banknotes, employing advanced image data processing techniques to effectively classify banknotes as genuine or counterfeit. The classification process is based on various features extracted from the images.

To achieve this goal, several classification algorithms, including Decision Tree, k-nearest neighbors (KNN), Random Forest, and Support Vector Machine (SVM), will be implemented and evaluated. Each algorithm will be trained and tested using a banknote authentication dataset containing data from images of both genuine and counterfeit banknotes. The project aims to compare the performance and accuracy of these algorithms to determine the most effective approach for banknote authentication.

The significance of this project lies in its potential to contribute to the security and integrity of financial transactions. By accurately identifying counterfeit banknotes, the proposed system aims to minimize financial losses and protect individuals, businesses, and financial institutions from the detrimental effects of counterfeit currency. Furthermore, the project seeks to enhance the efficiency of banknote authentication processes, reducing reliance on manual inspection and increasing the speed and accuracy of the authentication process. Ultimately, this research strives to strengthen the defenses against counterfeit currency, contributing to the overall stability of global economies.

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ACCRONYMN AND ABBREVIATION

RGB - Red, Green, Blue (referring to color channels in images).

SVM: -Support Vector Machine (a classification algorithm).

OCR: -Optical Character Recognition (for detecting text on banknotes).

RFID: -Radio-Frequency Identification (technology for tracking and identifying banknotes).

HSV: -Hue, Saturation, Value (color space representation).

NN: - Neural Network (fundamental in many machine learning approaches).

KNN: - k-Nearest Neighbors (classification algorithm based on proximity).

ISO- - International Standard For Organization

BAS- Banknote Authentication System

PCA- Principal Component Analysis

HOG - Histogram of Oriented Gradients

SIFTS - cale-Invariant Feature Transform

SURF Speeded Up Robust Features

ROC- Receiver Operating characteristic

CHAPTER ONE

INTRODUCTION

Counterfeit banknotes pose a significant threat to the financial systems and economies worldwide. Despite the increasing digitalization of transactions, physical currency in the form of banknotes continues to be widely used. With advancements in printing technology, it has become easier for counterfeiters to produce fake banknotes that closely resemble genuine ones, making it extremely challenging to differentiate between the two visually. This has led to an urgent need for reliable and efficient banknote authentication systems that can accurately identify counterfeit banknotes.

The primary objective of this project is to address this challenge by leveraging classification algorithms to enhance banknote authentication. By utilizing image data extracted from banknotes, the project aims to develop a robust system that can effectively classify banknotes as genuine or counterfeit. This classification process will be based on various features extracted from the images using advanced image processing techniques.

To achieve this objective, several classification algorithms will be implemented and evaluated.

These algorithms include Decision Tree, k-nearest neighbors (KNN), Random Forest, and Support Vector Machine (SVM). Each algorithm will be trained and tested using a banknote authentication dataset that contains data from s of genuine and counterfeit banknotes. The performance and accuracy of each algorithm will be compared to determine the most effective and reliable approach for banknote authentication.

The significance of this project lies in its potential to contribute to the security and integrity of financial transactions. By accurately identifying counterfeit banknotes, the proposed system can minimize financial losses and protect individuals, businesses, and financial institutions from the detrimental effects of counterfeit currency. Furthermore, the project aims to enhance the efficiency of banknote authentication processes, reducing the reliance on manual inspection and increasing the speed and accuracy of the authentication process.

This project aligns with the growing importance of utilizing machine learning and artificial intelligence techniques in various domains. By applying classification algorithms to the field of banknote authentication, it demonstrates the practical application of these technologies in real world scenarios. Moreover, the project offers valuable insights into the performance and accuracy of different classification algorithms in the context of banknote authentication, aiding in the selection and implementation of the most suitable algorithm for practical deployment.

In conclusion, this project aims to develop a reliable and efficient banknote authentication system by leveraging classification algorithms.. The subsequent chapters of this documentation will delve into the dataset, architecture, experimental methodology, results, and conclusion of the project, providing a comprehensive understanding of the approach and outcomes.

1.1 Background of the Study

Counterfeiting of banknotes poses a serious threat to financial security, necessitating advanced and efficient banknote authentication systems. Machine learning algorithms, particularly Decision Tree, KNN, Random Forest, and SVM, offer promising solutions for accurate classification of genuine and counterfeit banknotes. However, comprehensive research comparing these algorithms specifically for banknote authentication is limited.

This project aims to develop an enhanced banknote authentication system utilizing classification algorithms and evaluating their performance to identify the most effective one. The study will focus on digital image data analysis and consider dataset availability, feature extraction, and real world scenarios. The outcomes will benefit financial institutions, cash-handling businesses, and government agencies combatting counterfeiting threats. Moreover, the insights gained can have broader applications in pattern recognition and decision-making tasks beyond banknote authentication.

1.2 Problem Statement

The topic of banknote authentication is a complicated problem at the intersection of money, security, and technology. In an increasingly interconnected and computerized global economy, reliable and fast methods for distinguishing genuine banknotes from counterfeit ones have become critical. Counterfeiting, an age-old illegal activity, has grown in complexity as digital printing technologies have advanced, necessitating the development and deployment of effective authentication solutions. This problem statement aims to clarify the complexities and imperatives surrounding banknote authentication, evaluating its significance in the context of financial stability, technological innovation, regulatory compliance and development of a model that will aid in bank note authentication.

The need for banknote authentication stems from a number of interconnected dimensions and driving factors. The constant threat of counterfeit cash, first and foremost, threatens the integrity of financial systems and erodes trust in monetary transactions. To keep ahead of these adversaries, the continuous progress of counterfeiters' strategies, enabled by digital imaging and printing technology, necessitates a commensurate evolution in authentication systems. Furthermore, governments and central banks constantly improve currency security, demanding equivalent advancements in authentication methods. The global circulation of currency, aided by international trade and tourism, need universal and adaptive authentication systems.

Technology is crucial in banknote authenticity. With the advent of powerful image data processing algorithms, machine learning models, and specialized sensors, very accurate and efficient authentication systems have been developed. These systems, which are frequently integrated into ATMs, banknote counters, and point-of-sale devices, are critical to the integrity of financial transactions. Regulatory organizations, in addition to technological improvements, require the adoption of authentication technology to ensure monetary system stability and counterfeit deterrence.

Finally, the project aims to contribute to the development of best practices and novel banknote authentication technologies, addressing the multifaceted issues faced by counterfeiters while boosting trust and security in the global financial landscape.

1.3 Research Questions

To address the problem at hand, the following research questions will guide this project:

What features can be extracted from banknote images to differentiate between genuine and counterfeit banknotes?

1. Which classification algorithms, such as Decision Tree, KNN, Random Forest, and SVM, are most effective in accurately classifying banknotes?
2. How can the accuracy of these algorithms be compared to identify the best approach for banknote authentication?
3. What is the impact of dataset size and composition on the performance of banknote authentication algorithms?

1.4 Objectives

1.4.1 General Objectives

To develop an enhanced banknote authentication system by exploring and comparing different classification algorithms (Decision Tree, KNN, Random Forest, SVM) to identify the most effective one for accurately distinguishing between genuine and counterfeit banknotes. The deliverables of this project are as follows:

1: Identify discriminative features from banknote images for genuine and counterfeit differentiation.

2: Compare Decision Tree, KNN, Random Forest, and SVM algorithms for banknote classification.

3: Determine the best algorithm for accurate banknote authentication through performance evaluation.

4: Assessing the Impact of Dataset Size and Composition on Algorithm Performance.

1.5 Justification

The detection and prevention of counterfeit banknotes are of paramount importance to maintain the integrity of financial systems. By implementing an automated banknote authentication system based on classification algorithms, the accuracy and efficiency of the authentication process can be significantly enhanced. This project aims to contribute to the development of such a system, providing a reliable solution to the persistent problem of counterfeit banknotes.

1.6 Scope and Limitations of the Project

This project focuses on the development and evaluation of classification algorithms for banknote authentication. The scope includes implementing and analysing the performance of Decision Tree, KNN, Random Forest, and SVM algorithms using the banknote authentication dataset. The project aims to extract relevant features from banknote images and use these features as inputs for the classification algorithms. The primary focus is on the digital analysis of banknote images rather than physical inspection. The project does not involve the production or modification of banknotes. However , it is important to acknowledge certain limitations and constraints that may affect the scope of the project:

1. Data Limitation: Classification algorithm accuracy may be influenced by dataset availability, size, and diversity, limiting generalizability to all banknote types.

2. Feature Limitation: Extracted image data features are assumed sufficient for accurate classification, excluding factors like texture or ink quality that may affect banknote authenticity.

3. External Factors: Variations in lighting, image quality, and banknote designs are not considered, potentially impacting the system's accuracy in real-world scenarios.

4. Algorithm Selection: Limited to Decision Tree, KNN, Random Forest, and SVM algorithms, other machine learning approaches are unexplored, potentially affecting performance.

5. Real-time Implementation: Not involving real-time systems or integration with existing infrastructure, limited to analysis and evaluation using a predetermined dataset. Practical implementation and deployment are beyond the project's scope.

CHAPTER TWO

LITERATURE REVIEW

2.0 Introduction

Modern financial and economic systems heavily rely on the field of banknote authenticity. For safeguarding the integrity of financial transactions, limiting the circulation of counterfeit money, and ensuring trust and confidence in the currency supply chain, reliable banknote identification and verification are crucial. This chapter takes in-depth review of the literature on banknote authentication, security, and administration.

2.1 Banknote Authentication

The concept of money has been essential to human civilization, aiding trade, business, and financial dealings. Printed money (Poon et al., 2020), known as banknotes, is the type of money most commonly used in the modern-world. Any sound and functional economy is built on the issuing of banknotes by central banks and other financial institutions, which also supports trust and confidence in the financial system. The extensive usage of banknotes, however, also makes them a target for counterfeiting, necessitating the need for reliable banknote authentication systems to guarantee the accuracy of financial transactions.

The problem of fake currency is not new, and throughout history, counterfeiters have come up with inventive ways to imitate the security measures of real currency (Kumar & Prakash, 2012). Traditional methods of visual inspection have become inadequate as a result of the sophistication of counterfeit tactics, which has increased along with technology. The creation and use of sophisticated banknote authentication systems has resulted from this.

In this era of rapid global trade and technological advancement, effective banknote authentication is more important than ever (Zhang et al., 2021). In addition to safeguarding the integrity of the money supply chain (Kumar & Prakash, 2012), the validity of banknotes is crucial for defending the interests of people, companies, and governments. Currency fraud can result in large financial losses, damage public confidence in the financial system, and support illicit activities like tax evasion and money laundering (Li & Jain, 2011).

According to The Royal Swedish Academy of Sciences, "The importance of secure and reliable banknote authentication cannot be overstated, as it is the linchpin of trust in modern economies" (Academy of Sciences, 2020). It is therefore crucial to comprehend the changing nature of banknote authentication and investigate the impact of new technologies, laws, and standards on the development of money security.

As the physical representation of a nation's money, banknotes play a crucial part in contemporary economies. Numerous transactions are made possible by them, they act as a store of wealth, and they represent the strength and power of the issuing country. The significance of verifying the legitimacy of banknotes in this situation cannot be emphasized. A crucial tenet of monetary safety and economic stability is banknote authentication, the act

of confirming the legality of paper money. The necessity to distinguish between real banknotes and their counterfeit counterparts has sparked technological advancements, sophisticated verification systems, and the development of advanced security measures (Zhang et al., 2021).

As currency counterfeiters continue to refine their techniques, the task of authentication becomes increasingly complex. In order to comprehend the most recent developments, difficulties, and new trends in banknote authentication, it is crucial to examine the level of knowledge and research in this area today. In-depth analysis of these subjects is the goal of the literature review, which also lays the groundwork for future development and improvement of banknote authentication management systems (Li & Jain, 2011).

2.1.1 Early Currency and Authentication Challenges

Banknotes have been used as a form of exchange since the dawn of time (Poon et al., 2020). Promissory notes, which were used by merchants during the Tang Dynasty in 7thcentury China were the earliest examples of paper money. European banks didn't start issuing the first contemporary banknotes as a form of representational money, however, that could be exchanged for a specific amount of precious metal on demand until the late 17th century (Möser et al., 2017).

Counterfeiting has existed almost as long as banknotes themselves. Counterfeiters in earlier centuries relied on techniques like engraving and printing to replicate banknotes (Zhang et al., 2021). These counterfeit currencies threatened both the credibility of banks and the financial stability of nations. To combat counterfeiting, historical banknotes incorporated a range of security features. Watermarks, which date back to the 13th century, were one of the earliest security features (Möser et al., 2017). They were often created by variations in paper thickness or the addition of threads during the

papermaking process. Additionally, complex engraving, intricate designs, and unique serial numbers were introduced to deter counterfeiters (Zhang et al., 2021).

2.1.2 Emergence of Security Features

Banknote security elements have been evolving over time, influenced by the problems provided by counterfeiters for millennia. The security features included into banknotes evolved along with counterfeiting methods. This section explores the historical development

of banknote security features, highlighting significant advancements that have aided in the advancement of banknote authentication.

2.1.2.1 Watermarks and Threads

One of the earliest security features used in banknotes was the watermark. Watermarks, which date back to the 13th century, were initially created by variations in paper thickness (Li & Jain, 2011).

These faint designs, visible when held against the light, served as a basic but effective anticounterfeiting measure. In addition to watermarks, security threads, often made of silk or metal, were sometimes woven into the paper, making it challenging for counterfeiters to reproduce the banknote's texture and appearance (Zhang et al., 2021).

2.1.2.2 Engraving and Intricate Design

Banks and governments used sophisticated engravings and designs to dissuade counterfeiters as the use of banknotes expanded in the 17th and 18th centuries. Intricate and complicated patterns, which were costly and time-consuming to replicate, were made by skilled engravers (Möser et al., 2017). These creative components provided both security and aesthetic purposes, making counterfeiting more difficult.

2.1.2.3 Serial Numbers and Unique Identifiers

Another important security improvement was the addition of serial numbers on banknotes. An individual serial number was given to each banknote, allowing for the tracking and identification of specific notes (Li & Jain, 2011). Due to the ease with which forgeries serial numbers could be identified, this feature not only made recordkeeping easier but also served to discourage counterfeiters.

2.1.2.4 Modern Security Features

The 20th century marked a turning point in banknote security with the emergence of modern technologies. Holograms, for example, added a highly sophisticated security layer. When

incorporated into banknotes, holograms created an ever-changing, three-dimensional image that was virtually impossible for counterfeiters to replicate. Additionally, security threads with microprinting and colour-shifting ink became common, further complicating counterfeiting efforts (Möser et al., 2017).

2.1.2.5 Incorporation of Transparent Windows and Complex Patterns

In recent decades, banknotes have integrated transparent windows with intricate designs. These features are not only aesthetically appealing but also incredibly challenging to reproduce accurately. The use of complex patterns, which change when viewed from different angles, provides an additional level of security.

2.1.2.6 Innovations in the 21st Century

In the digital age, banknote security features have continued to evolve. Advancements include machine-readable features such as barcodes and RFID tags, which are used in modern authentication systems to facilitate rapid and accurate verification. The emergence of these security features reflects a continual effort to stay ahead of counterfeiters and maintain trust in the integrity of currency (Li & Jain, 2011). The historical progression from watermarks and engravings to modern technologies demonstrates the resilience of banknote authentication in the face of evolving threats. Understanding this history is crucial in appreciating the complexity of contemporary banknote authentication systems (Zhang et al., 2021).

2.2 Banknote Security Features

Banknote security features are the critical elements integrated into currency design to deter counterfeiting and ensure the authenticity of banknotes. Over the centuries, a wide range of security features has been developed and incorporated into banknotes to safeguard against counterfeit practices. This section provides an overview of the most common and advanced security features used in banknotes, and their role in enhancing banknote authentication.

2.2.1 Watermarks

One of the earliest security features, watermarks, have been utilized for centuries. Watermarks involve creating subtle designs or text in the paper during the manufacturing process, making

them visible when held up to light. These designs serve as both an aesthetic element and a security feature, helping to verify the authenticity of the banknote (Fabri, 2016).

2.2.2 Security Threads

Security threads are thin strips of material, such as silk or metal, embedded within the paper during banknote production. They may feature holographic or microprinted elements and are challenging to replicate. Security threads enhance banknote authentication by their presence and unique characteristics (Beck & Gagliardini, 2018).

2.2.3 Holograms

Holograms are 3D images that are integrated into banknotes. These images change when viewed from different angles, making it extremely difficult for counterfeiters to reproduce. Holograms are both visually striking and highly effective security features (Bank of England, 2016).

2.2.4 Microprinting

Microprinting involves printing tiny, intricate text or patterns on banknotes, often invisible to the naked eye without magnification. These microprints are difficult to replicate with standard printing methods and act as a powerful deterrent to counterfeiters (Adams, 2012).

2.2.5 Color-Shifting Ink

Colour-shifting ink is used to print certain elements on banknotes, such as denominations or symbols. When viewed from different angles, the ink changes color, making it challenging to replicate accurately. This feature provides both a visual cue for authentication and an added layer of security (European Central Bank, 2021).

2.2.6 Transparent Windows

Many modern banknotes incorporate transparent windows with complex and intricate designs. These windows are not only visually appealing but also incredibly challenging to reproduce. They enhance security while adding an aesthetic dimension to the banknote (Reserve Bank of Australia, 2018).

2.2.7 Machine-Readable Features

In the digital age, banknote security has extended to machine-readable features. These include barcodes and RFID tags, which are used for rapid and accurate banknote verification in automated authentication systems (Bank of Canada, 2019). The use of these security features reflects the ongoing efforts to stay ahead of counterfeiters and maintain trust in the authenticity of banknotes. Understanding these features is crucial in comprehending the complexity of contemporary banknote authentication systems.

2.3 Banknote Counterfeiting

Banknote counterfeiting has been a persistent challenge throughout the history of currency. Counterfeiters have employed various techniques to replicate banknotes and deceive the public and financial institutions. This section provides an overview of the historical and contemporary methods of banknote counterfeiting, as well as statistics and research related to this issue.

Historically, counterfeiting methods involved techniques such as engraving, intaglio printing, and manual reproduction (Burke, 2015). Counterfeiters painstakingly replicated banknotes, often with varying degrees of success. The introduction of security features in the 17th and 18th centuries prompted counterfeiters to become more sophisticated in their techniques (Cuhaj & Brandimore, 2013).

In the digital age, modern counterfeiters have access to advanced printing technology, making it easier to produce high-quality counterfeit banknotes (Gao, 2006). Digital scanning, color laser printing, and advanced graphic design software enable counterfeiters to create convincing replicas (Cowell, 2018). Statistics on banknote counterfeiting provide a clear picture of the scale of the problem. For instance, in the United States, the United States Secret Service reported that approximately $6.7 million in counterfeit currency was seized in 2020 (United States Secret Service, 2021). These figures highlight the ongoing threat posed by counterfeit currency.

Counterfeiting is not merely a technical problem but also an economic issue. Leal and Taylor (2015) found that the presence of counterfeit currency in circulation could have inflationary effects, impacting a nation's economic stability. Efforts to combat counterfeiting have led to significant research in the development of banknote authentication systems and anti-counterfeiting technologies. For example, researchers have explored the effectiveness of ultraviolet (UV) and infrared (IR) detection methods in identifying counterfeit banknotes (Zhang et al., 2019). The development of machine learning and artificial intelligence in counterfeit detection has also been a focus of research (Poon et al., 2020).

2.4 Authentication Technologies

Authentication technologies are the cornerstone of banknote security, offering diverse methods for verifying the legitimacy of banknotes. These technologies have evolved significantly over time, leveraging innovations in materials science, optics, and electronics to combat counterfeiting more effectively. This section explores various authentication technologies and their role in ensuring the integrity of currency.

2.4.1 Ultraviolet (UV) Detection

Ultraviolet (UV) detection is a well-established technology used in banknote authentication. UV detectors emit ultraviolet light onto a banknote, revealing hidden features that are invisible to the naked eye. UV-activated inks and patterns are commonly incorporated into banknotes

as security features, making this technology a fundamental tool for authentication (Goodhew, 2019).

2.4.2 Infrared (IR) Detection

Infrared (IR) detection is another prevalent technology in banknote authentication. IR detectors capture the infrared spectrum of light, revealing unique patterns and features that counterfeit banknotes often lack. This technology is particularly effective in detecting differences in ink types and densities (Adams, 2012).

2.4.3 Magnetic Ink Detection

Magnetic ink detection involves scanning banknotes for the presence of magnetic ink. Magnetic ink is used in various security features, including serial numbers and security threads. This technology allows for quick and accurate authentication, as counterfeit banknotes typically lack genuine magnetic ink elements (Beck & Gagliardini, 2018).

2.4.4 Machine-Readable Features

Banknotes are increasingly integrating machine-readable features, such as barcodes and RadioFrequency Identification (RFID) tags. These features enable automated authentication systems to quickly and accurately verify the authenticity of banknotes. Barcodes, for example, can store information related to the banknote's origin and security features (Bank of Canada, 2019).

2.4.5 Visible Features and Microprinting

Visible features, including intricate designs and microprinting, are vital for manual banknote authentication. Microprinting involves the use of tiny, intricately detailed text or patterns that

are difficult to reproduce with standard printing methods. These features serve as visible cues for authenticity (European Central Bank, 2021).

2.4.6 Digital Imaging and Scanning

Advancements in digital imaging and scanning technologies have enhanced banknote authentication. High-resolution scanners and digital imaging systems can capture even the most subtle security features and reveal anomalies that counterfeit banknotes may possess (Smith & Johnson, 2014).

2.4.7 Multi-Spectral Imaging

Multi-spectral imaging combines various light spectra, including UV, IR, and visible light, to capture detailed information from banknotes. This technology can reveal a wide range of security features and provide a comprehensive assessment of a banknote's authenticity (Laramee et al., 2015). These authentication technologies collectively contribute to the robustness of banknote authentication systems. By combining both visible and hidden security elements, they offer an effective defense against counterfeiting, bolstering trust in the currency supply chain and financial transactions.

2.5 Banknote Authentication Systems

Banknote authentication systems are integral to the safeguarding of financial transactions and the prevention of counterfeit currency circulation. These systems employ a combination of technologies, algorithms, and security features to verify the authenticity of banknotes. There exist various banknote authentication systems that play an important role in maintaining trust in currency.

2.5.1 Manual Authentication Systems

Manual banknote authentication methods involve human inspection of physical security features, such as watermarks, security threads, and holograms. These methods require training

and expertise, and their effectiveness relies on the knowledge and attention to detail of the user (Smith & Johnson, 2014).

2.5.2 Automatic Authentication Systems

Automatic banknote authentication systems have gained prominence in recent years. They employ a combination of sensors, imaging devices, and machine-readable features to quickly and accurately verify banknote authenticity. Such systems are used in various industries, including banking, retail, and vending, to prevent counterfeit banknotes from entering circulation (Beck & Gagliardini, 2018).

2.5.3 Ultraviolet (UV) and Infrared (IR) Scanners

UV and IR scanners are common components of automatic banknote authentication systems. They are used to detect UV- and IR-activated security features on banknotes. These scanners offer rapid authentication and are highly effective in identifying counterfeit banknotes (Goodhew, 2019).

2.5.4 Magnetic Ink Detection Systems

Magnetic ink detection systems use sensors to check for the presence of magnetic ink in banknotes. They can swiftly determine whether the magnetic ink used in elements such as serial numbers and security threads is genuine, helping to identify counterfeit notes (Bank of England, 2016).

2.5.5 Machine-Readable Features and Barcode Scanners

Banknotes with machine-readable features, such as barcodes or RFID tags, can be authenticated using dedicated scanners. These scanners quickly extract data from the banknote to verify its authenticity, making them essential in automated systems (Bank of Canada, 2019).

2.5.6 Artificial Intelligence (AI) and Machine Learning

AI and machine learning algorithms have been increasingly employed in banknote authentication systems. These technologies can learn from historical data and adapt to new counterfeit techniques, improving the accuracy of authentication (Laramee et al., 2015).

2.5.7 Multi-Spectral Imaging Systems

Multi-spectral imaging systems capture banknote images under various light spectra, enabling a comprehensive analysis of security features. They can identify a wide range of security elements, including watermarks, microprinting, and colour-shifting ink (European Central Bank, 2021). These banknote authentication systems are critical for preventing counterfeit banknotes from entering circulation, ensuring the trust and integrity of financial transactions, and maintaining the stability of economies.

2.6 Case Studies on Existing Banknote Authentication Systems

Examining real-world examples of banknote authentication systems provides valuable insights into the effectiveness, challenges, and best practices in ensuring the security and authenticity of currency.

2.6.1 Euro Banknote Authentication Systems

The European Central Bank (ECB) has been at the forefront of developing advanced banknote authentication systems for the Euro currency. These systems incorporate a variety of security features, including holograms, colour-changing ink, and ultraviolet and infrared elements,

making them robust and resilient against counterfeiting attempts (European Central Bank, 2021).

2.6.2 Swiss Franc Authentication

The Swiss National Bank (SNB) introduced new Swiss banknotes with enhanced security features, such as a transparent window and microprinting. The SNB's approach includes an extensive public awareness campaign, educational materials, and collaboration with financial institutions to ensure the successful adoption of these features by the public (Swiss National Bank, 2020).

2.6.3 Canadian Currency Authentication

Canada's central bank, the Bank of Canada, introduced new polymer banknotes with advanced security elements, including holograms and transparent windows.
Additionally, the Bank of Canada collaborated with stakeholders, including retailers and law enforcement, to educate the public about the new security features and provide training for banknote authentication (Bank of Canada, 2019).

2.6.4 Australian Banknote Verification

The Reserve Bank of Australia introduced a new series of banknotes featuring innovative security elements such as top-to-bottom clear windows. The Reserve Bank actively engages with banks and financial institutions to ensure the efficient verification of these banknotes using automated banknote authentication systems (Reserve Bank of Australia, 2018).

2.6.5 United States Currency Authentication

The U.S. Bureau of Engraving and Printing, in collaboration with the U.S. Secret Service,

continually improves security features on U.S. banknotes. These include color-shifting ink, microprinting, and watermarks. To assist businesses and the public in authenticating banknotes, the U.S. government offers educational resources and training (U.S. Secret Service, 2021).

2.7 Technology and Banknotes Authentication

2.7.1 Biometric Authentication in Banking

Biometric authentication is a cutting-edge technology that has gained significant prominence in the banking sector as a means to enhance security, improve customer experience, and reduce the risk of fraud.

2.7.1.1 Biometric Technologies in Banking

Fingerprint Recognition: Many banks have adopted fingerprint recognition as a common biometric authentication method. Customers can use their registered fingerprints to access their accounts, make transactions, and verify their identity securely (Adams, 2012).

Face Recognition: Facial recognition technology has become increasingly prevalent in the banking industry. Customers can log into their accounts or conduct transactions by simply showing their face to a camera or mobile device equipped with facial recognition software (Li & Jain, 2011).

Voice Biometrics: Voice recognition technology is employed by some banks to verify a customer's identity based on their unique vocal characteristics. Customers can perform transactions by speaking to a system that analyzes their voice for authentication (Aloul et al., 2011).

2.7.1.2 Security and Convenience

Biometric authentication in banking offers a high level of security. Unlike traditional methods such as passwords or PINs, biometric characteristics are unique to each individual and difficult to replicate. This reduces the risk of unauthorized access to bank accounts. Additionally,

biometric authentication is convenient for customers as it eliminates the need to remember and enter complex passwords (Jain et al., 2016).

2.7.1.3 Applications in Online and Mobile Banking

Biometric authentication is commonly used in online and mobile banking applications. Customers can log in to their accounts using their biometric data, such as fingerprints or facial recognition, via smartphones and tablets. This not only enhances security but also provides a seamless and userfriendly experience (Ratha et al., 2001).

2.7.1.4 Challenges and Privacy Concerns

Despite the advantages of biometric authentication, there are challenges and privacy concerns. Storage and protection of biometric data, as well as potential breaches, raise significant privacy issues. It is crucial for banks to implement robust security measures to safeguard biometric data and ensure customer privacy (Ratha et al., 2001).

2.7.1.5 Compliance and Regulations

Financial institutions must also comply with regulatory standards, such as the General Data Protection Regulation (GDPR) in Europe, which mandates how biometric data should be handled, stored, and protected. Meeting these requirements is essential to avoid legal issues (Krol, 2018). The integration of biometric authentication in banking underscores the industry's commitment to enhancing security and customer convenience. As technology continues to advance, biometric solutions will play an increasingly significant role in securing financial transactions and ensuring the protection of customer data.

2.7.2 Blockchain and Cryptocurrency Impact

Blockchain technology and cryptocurrencies have revolutionized the financial industry, offering new opportunities and challenges for banks and financial institutions.

2.7.2.1 Decentralization and Trust-less Transactions

Blockchain, the underlying technology of cryptocurrencies like Bitcoin, offers a decentralized ledger that eliminates the need for intermediaries. Banks are exploring how blockchain can streamline cross-border transactions, reduce settlement times, and enhance transparency (Mougayar, 2016). Consequently, Cryptocurrencies allow for trustless transactions, where parties can transact directly without relying on banks or other intermediaries. This can lead to lower transaction costs and quicker settlement times.

2.7.2.2 Digital Currencies and Central Bank Digital Currencies (CBDCs)

Central banks are investigating blockchain technology for the development of Central Bank Digital Currencies (CBDCs). These digital currencies, issued and regulated by central banks, aim to enhance the efficiency of payment systems and reduce the risks associated with traditional paper currency (Bank for International Settlements, 2020). Additionally, the rise of cryptocurrencies has prompted central banks to consider the development of their own digital currencies to maintain control over monetary policy and protect financial stability (European Central Bank, 2019).

2.7.2.3 Enhanced Security and Anti-Fraud Measures

Blockchain's inherent security features make it difficult for fraudulent activities like double spending. Banks are exploring the use of blockchain to enhance security in areas such as identity verification, asset management, and document verification (Tapscott & Tapscott, 2016). Along this, the transparent and immutable nature of cryptocurrency transactions can reduce the risk of fraud and counterfeiting in financial transactions.

2.7.2.4 Financial Inclusion

Blockchain can facilitate financial inclusion by providing access to banking services for underserved populations. This technology can help reduce the costs of financial services and enable secure cross-border remittances (Swan, 2015). Cryptocurrencies can enable individuals in regions with limited access to traditional banking services to participate in the global economy, potentially reducing the barriers to financial inclusion.

2.7.2.5 Regulatory and Compliance Challenges

The decentralized nature of blockchain and cryptocurrencies poses challenges for regulatory authorities. Banks and financial institutions are navigating the regulatory landscape to ensure compliance with Anti-Money Laundering (AML) and Know Your Customer (KYC) requirements (Yermack, 2018). Cryptocurrencies on the other hand, have prompted regulatory bodies to develop frameworks to manage risks associated with their use, ensuring consumer protection and compliance with existing financial regulations. The impact of blockchain and cryptocurrencies in banking continues to evolve, offering opportunities for innovation and efficiency while requiring careful consideration of regulatory and security implications. As these technologies mature, their role in reshaping the financial industry is likely to become increasingly significant.

2.7.3 Cryptocurrency Authentication and Security Considerations

The use of cryptocurrencies requires robust authentication and security measures to protect digital assets from theft, fraud, and unauthorized access. The key measures include: -

2.7.3.1 Private Key Management

Cryptocurrency wallets use private keys to authenticate ownership and facilitate transactions. Users must safeguard their private keys, which serve as digital signatures. Authentication involves the secure storage and management of these keys (Möser et al., 2017). Users should keep their private keys offline, use hardware wallets, or employ secure, encrypted storage solutions. Multisignature wallets, which require multiple private keys for transactions, enhance security (Eyal et al., 2016).

2.7.3.2 Two-Factor Authentication (2FA)

Two-factor authentication adds an extra layer of security to cryptocurrency accounts. It typically involves something the user knows (password) and something the user possesses (a device or code). This dual authentication provides an additional barrier to unauthorized access (Korol et al.,
2019). Cryptocurrency exchanges and wallet services often offer 2FA options. Users should enable 2FA to reduce the risk of unauthorized account access.

2.7.3.3 Cold Storage and Hardware Wallets

Cold storage involves keeping cryptocurrency holdings offline, making them inaccessible to online threats. Hardware wallets are physical devices that store private keys offline and provide an additional layer of security (Tian et al., 2018). Many investors opt for cold storage and hardware wallets to protect their cryptocurrency holdings from online vulnerabilities and hacking attempts (Zohar & Teutsch, 2015).

2.7.3.4 Secure Backup and Recovery

In case of loss, theft, or damage to authentication methods (such as private keys), users should have secure backup and recovery procedures in place. This ensures access to their cryptocurrency holdings (Möser et al., 2017). Users should store backup information securely, preferably in multiple physical locations, and follow best practices for recovery seed management.

2.7.3.5 Ongoing Vigilance Against Phishing and Scams

Cryptocurrency users should exercise caution to avoid falling victim to phishing attacks, scams, and fraudulent schemes. Such attacks often attempt to trick users into revealing private keys or other sensitive information (Conti & Sobieski, 2019). Education and awareness are critical. Users should verify the authenticity of websites, emails, and communication channels, and never share sensitive information with unknown or unverified sources.

2.7.3.6 Regulatory Compliance

Cryptocurrency exchanges and services often have regulatory requirements for user authentication, such as Know Your Customer (KYC) and Anti-Money Laundering (AML) procedures. Users should be prepared to comply with these regulations (Yermack, 2018). Compliance with regulations is essential for the security and legitimacy of cryptocurrency activities. Users should be aware of the requirements and ensure the platforms they use follow best practices in compliance.

2.7.3.7 Continuous Monitoring and Updates

Staying informed about security threats and implementing updates and patches for wallet software and systems is crucial for ongoing authentication and security (Conti & Sobieski,

2019). Cryptocurrency users should regularly update their wallet software, use the latest security measures, and keep abreast of security threats and best practices. The security of cryptocurrencies relies heavily on the diligence of users in protecting their assets. By understanding and implementing these authentication and security considerations, individuals can enhance the protection of their cryptocurrency holdings.

2.7.4 Machine Learning and AI in Authentication

Machine learning and artificial intelligence (AI) are increasingly utilized in authentication systems to enhance security, reduce fraud, and improve user experience. The key concepts include:-

2.7.4.1 Behavioral Biometrics

Behavioral biometrics use machine learning to analyze user behavior, such as typing patterns, mouse movements, and touchscreen gestures, to create unique user profiles. This enables continuous authentication by identifying unusual behavior (Monrose et al., 2017). Behavioral biometrics offer a non-intrusive authentication method that adapts to the user's behavior. AI can detect anomalies, signaling potential security threats.

2.7.4.2 Voice and Speech Recognition

Voice and speech recognition powered by AI can verify a user's identity based on their voice characteristics. This is used for phone banking, call centers, and voice-activated assistants (Baker et al., 2014). Voice recognition can be a convenient and secure method for authentication. AI algorithms can detect voice spoofing attempts and ensure the authenticity of the speaker.

2.7.4.3 Facial Recognition

Facial recognition technology utilizes AI to analyze facial features and compare them to stored biometric data. It is employed for access control, mobile device authentication, and identity verification (Li & Jain, 2011). AI-driven facial recognition can enhance security by quickly verifying identity. It should incorporate anti-spoofing measures to protect against fraudulent attempts.

2.7.4.4 Predictive Analytics and Anomaly Detection

Machine learning algorithms can assess user behavior, transaction history, and other data to create predictive models. Anomalies can be detected, flagging potentially fraudulent activities (Kaur et al., 2016). Predictive analytics and anomaly detection can enhance security by identifying unusual activities. AI continuously learns from data to improve detection accuracy.

2.7.4.5 Natural Language Processing (NLP)

NLP, a branch of AI, is used in chatbots and virtual assistants for user authentication through natural language conversations. Users can ask security questions or provide spoken passphrases (Zhang et al., 2021). AI-driven NLP enhances user experience and can securely authenticate users through voice or text interactions.

2.7.4.6 Multi-Modal Authentication

Multi-modal authentication combines various AI-powered biometric methods, such as face recognition, voice verification, and fingerprint scanning, to enhance security and reduce the risk of false positives (Möser et al., 2017). Combining multiple biometric methods increases authentication accuracy while also offering redundancy in case one method fails or is compromised.

2.7.4.7 Continuous Authentication

AI-driven continuous authentication constantly monitors user behavior and interactions with systems. If anomalies or threats are detected, access can be revoked or additional verification steps initiated (Monteiro et al., 2017). Continuous authentication powered by AI helps protect against unauthorized access and fraud by dynamically assessing risk and adjusting security measures.

AI and machine learning have transformed authentication by enabling intelligent, adaptive, and efficient methods that enhance security while providing a seamless user experience. As technology continues to advance, AI-driven authentication will play an increasingly vital role in protecting digital assets and sensitive data.

2.7.5 Machine Learning and AI in Authentication Algorithms

Machine learning and artificial intelligence (AI) have brought significant advancements to authentication algorithms, enabling sophisticated and adaptive security measures.

2.7.5.1 Anomaly Detection with Machine Learning

One-Class SVM (Support Vector Machine), Random Forest, or Autoencoders are used to create models of normal user behavior. When deviations or anomalies occur, the system can trigger additional authentication or block access (Kaur et al., 2016). Anomaly detection is applied to continuous authentication, identifying unusual user activity, and preventing unauthorized access in real-time.

2.7.5.2 Behavioral Biometric Authentication

Machine learning models, often based on deep learning techniques, analyze user behavior, such as typing patterns, mouse movements, and touchscreen gestures. These models create unique user profiles for authentication (Monrose et al., 2017). Behavioral biometrics are employed for continuous authentication, improving security by continuously monitoring user behavior.

2.7.5.3 Facial Recognition with Deep Learning

Convolutional Neural Networks (CNNs) and Deep Learning algorithms are used for facial recognition. These algorithms extract facial features and match them against stored biometric data (Parkhi et al., 2015). Facial recognition is employed for access control, identity verification, and mobile device authentication.

2.7.5.4 Voice and Speech Verification with AI

Natural Language Processing (NLP) and deep learning algorithms analyze speech patterns and voice characteristics. These algorithms verify a user's identity based on their voice (Han et al.,

2015). Voice and speech verification is utilized in phone banking, call centers, and voice-activated assistants for user authentication.

2.7.5.5 Multi-Modal Authentication

Multi-modal authentication algorithms combine various biometric methods, such as facial recognition, voice verification, and fingerprint scanning. These algorithms assess multiple biometric factors for enhanced security (Kumar & Prakash, 2012). Multi-modal authentication enhances authentication accuracy and is used for access control in high security environments.

2.7.5.6 Natural Language Processing (NLP) for Text-Based Authentication

NLP algorithms analyze natural language text and user conversations to assess their authenticity.

AI-driven NLP can verify users based on their responses to security questions (Zhang et al., 2021). NLP is used in chatbots and virtual assistants for user authentication through text-based conversations.

2.7.5.7 Continuous Authentication with Machine Learning

Machine learning models, including Random Forest or Recurrent Neural Networks (RNN), continuously assess user behavior and interaction with systems. AI algorithms detect anomalies and adapt security measures (Monteiro et al., 2017). ontinuous authentication enhances user experience and security by dynamically assessing risk and ensuring ongoing protection. These machine learning and AI-driven authentication algorithms have revolutionized the way authentication is performed. By continuously user behavior and adapting to emerging threats, they offer a proactive approach to security, reducing the risk of unauthorized access and fraud.

2.7.5.8 Advantages of Machine Learning and AI in Authentication

Machine learning and AI enable advanced biometric authentication methods, such as behavioral biometrics and facial recognition, that significantly enhance security (Kaur et al., 2016). AI-driven authentication allows for continuous user behavior monitoring, adapting security measures in realtime to respond to emerging threats (Monteiro et al., 2017). Machine learning and AI make authentication more convenient, with technologies like voice recognition and chatbots improving the user experience (Zhang et al., 2021). AI enables the integration of multiple authentication methods, combining facial recognition, voice verification, and more to enhance security and reduce the risk of false positives (Kumar & Prakash, 2012).

2.8.5.9 Challenges of Machine Learning and AI in Authentication

The collection and storage of biometric data for AI-driven authentication can raise privacy concerns. Users may worry about the misuse or unauthorized access to their biometric information.

While AI can enhance security, it can also introduce vulnerabilities if not properly secured.

Adversaries may attempt to manipulate AI models or exploit weaknesses in the algorithms. Meeting regulatory requirements, such as GDPR and KYC regulations, can be complex when implementing AI-driven authentication methods (Yermack, 2018). The ethical implications of AI in authentication, including biases in facial recognition algorithms and the potential for discrimination, need to be addressed (Buolamwini & Gebru, 2018). Integrating AI-driven authentication into existing systems and ensuring scalability can be complex and may require substantial resources and expertise.

2.9 Security Protocols and Standards in Banknote Counterfeiting Prevention

The prevention of banknote counterfeiting is a paramount concern for central banks and monetary authorities. A range of security protocols and standards have been established to deter counterfeiters and safeguard the integrity of currency, some of the critical security measures include: -

2.9.1 Watermarks

Watermarks are intricate designs or patterns embedded into the banknote paper during the manufacturing process. They are visible when held up to the light and can be used to authenticate the banknote (European Central Bank, 2021).

2.9.2 Holograms

Holograms are three-dimensional images that change when tilted, making them difficult to reproduce. They are often used as security features on banknotes (Reserve Bank of Australia, 2018).

2.9.3 Colour-Shifting Ink

Colour-shifting ink is used to print specific design elements or denominations on banknotes. The color changes when viewed from different angles, making it challenging for counterfeiters to replicate (U.S. Secret Service, 2021).

2.9.4 Ultraviolet (UV) and Infrared (IR) Elements

UV and IR features are incorporated into banknotes. These elements are visible only under UV or IR light, allowing authentication through specialized detectors (European Central Bank, 2021).

2.9.5 Microprinting

Microprinting involves printing tiny, intricate text or patterns on banknotes that are difficult to replicate with standard printing methods. Counterfeit banknotes often lack the fine detail found in genuine currency (Bank of Canada, 2019).

2.9.6 Security Threads

Banknotes are embedded with thin, colored threads that can be partially visible on the surface.

They often contain microprinting or other features that are difficult to reproduce (Swiss National Bank, 2020).

2.9.7 Raised Printing and Tactile Features

Raised printing and tactile features, such as embossed patterns, make banknotes distinguishable by touch. Counterfeit notes often lack these tactile elements (Reserve Bank of Australia, 2018).

2.9.8 Substrate and Material Composition

Banknote paper is made with specific substrates and materials that are difficult to source and reproduce. Central banks keep these components secret and use specialized techniques to make them (European Central Bank, 2021). These security protocols and standards are essential in the fight against banknote counterfeiting. They make it difficult for counterfeiters to replicate the intricate features of genuine banknotes, providing an effective defense against fraudulent activities.

2.9.9 Analysis of Banknote Authentication System using Machine Learning Techniques

Preserving genuineness of higher denomination printed Banknotes is one of the critical issues. It has the major role in financial activities of every country (Chhotu and Anil, 2015). The study in

(Chhotu and Anil, 2015) evaluates different machine learning algorithms and concludes that Decision-Tree and MLP technique is best to classify a bank note. In (Eugen and Volker, 2014), the features of the banknote are extracted using Fast Wavelet Transforms. Later, one-against-all classification approach was employed that classifies the note into four different categories:

Genuine, High-Quality Forgery, Low-Quality Forgery, and Inappropriate ROI which resulted in 100% detection rate. Evaluation of SVM and BPN is done in (Thirunavukkarasu et.al., 2017) and [4] where BPN outperforms SVM.

In (Thirunavukkarasu et.al., 2017), BPN gives 10% more accuracy than SVM and is determined as the best classifier for predicting proteins sequence based on their compositions, whereas in(Fabri, 2016) and (Zhang et al., 2021) SVM outperforms BPN. The results depend on the dataset and the type of classification problem. The research implements a system for classifying Thai banknotes using neural networks. Firstly, images of notes are collected by a scanner which is saved as bitmap data. Features are extracted from this data and inputted to BPN for learning and recognition. In (Möser et al., 2017), a new method is proposed for banknote recognition; Probabilistic Neural Network (PNN) and 100% success rate is obtained.

The study in (Zhang et al., 2021) uses LVQ classifier for banknote recognition. The experiment has been applied to US dollars and can be used for another kind of banknotes. The paper (Smith & Johnson, 2014) presents a system for detecting counterfeit currency banknotes. The note is processed using camera and image is divided into parts using segmentation. Watermark histogram features are extracted from different segmentation region to match the watermark with Gandhi's portrait. The result is shown on the user interface. In a similar study in (Kumar & Prakash, 2012), the features are extracted from images by segmenting the image.

These features are fed to SVM classifier which then determines the authenticity of the note. A system has been proposed in (Zhang et al., 2021) for recognition of euro banknotes. The study shows that three-layered perceptron is proved to classify the banknotes into a certain class accurately by taking the banknote images as inputs. This model has been trained by the backpropagation method. After classification, a Radial Basis Function network is used for validation which rejects the invalid data. The model gives 100% acceptance rate of valid notes and 0% acceptance rate of the invalid banknotes.

2.9 ISO Standards Related to Banknote Authentication

The International Organization for Standardization (ISO) plays a critical role in setting international standards for various industries, including those related to banknote authentication. ISO standards help ensure consistency, quality, and interoperability in authentication systems and technologies.

2.9.1 ISO 16642:2017 - Financial Services - Requirements for Message Authentication using Symmetric Techniques

Standard Description: ISO 16642 specifies security requirements for authenticating messages in the financial industry, including those related to banknote processing and transactions.

2.9.2 ISO 18774:2018 - Security and Resilience - Authentication and Exchange of Symmetric Keys

Standard Description: ISO 18774 addresses key management and authentication techniques, which are crucial components in ensuring the security of banknote handling and processing systems.

2.9.3 ISO 20022 - Financial Services - Universal Financial Industry Message Scheme

Standard Description: ISO 20022 provides a standardized framework for exchanging financial messages, which can include authentication and authorization processes related to banknote handling and transactions.

2.9.4 ISO 24306:2007 - Security and Control - Biometric Verification of Identity

Standard Description: ISO 24306 outlines requirements for biometric verification systems, which are commonly used in banknote authentication and access control.

2.9.5 ISO 24617-1:2009 - Language Resource Management - Part 1: Character Annotation

Standard Description: ISO 24617-1 defines character annotation codes that can be used for security features, including those on banknotes, to ensure interoperability and international recognition.

2.9.6 ISO 26262 - Road Vehicles - Functional Safety

Standard Description: While primarily related to automotive safety, ISO 26262 can inform the development of secure and reliable components used in banknote processing machines.

2.9.7 ISO 29002-10:2013 - Financial Services - Security Management

Standard Description: ISO 29002-10 addresses security management for financial services, which includes measures applicable to banknote processing systems and authentication. ISO standards provide a framework for ensuring the reliability and security of banknote authentication systems, contributing to the overall integrity of financial transactions and the prevention of counterfeiting. These standards guide the development and implementation of authentication technologies in the financial industry.

2.10 Challenges and Vulnerabilities in Banknote Authentication

While security protocols and standards are essential, the field of banknote authentication faces ongoing challenges and vulnerabilities that must be addressed to maintain the integrity of currency systems.

2.10.1 Counterfeiting Techniques

Counterfeiters continuously develop new and sophisticated techniques to replicate security features, making it challenging for authentication systems to keep up (The Central Bank Counterfeit Deterrence Group, 2018).

2.10.2 Digital Counterfeiting

Advancements in digital printing and scanning technologies have made it easier to produce high quality counterfeit banknotes. Digital counterfeiting poses a significant threat (Europol, 2020).

2.10.3 Accessibility of Authentication Technologies

As authentication technologies become more accessible, including the availability of advanced software and hardware, they can also be misused by counterfeiters to create convincing fake banknotes (The Central Bank Counterfeit Deterrence Group, 2018).

2.10.4 Insider Threats

Insider threats from individuals within the banknote printing and processing industry can pose vulnerabilities. Unauthorized access or insider knowledge can be exploited by counterfeiters (Bazijanac et al., 2020).

2.10.5 Regulatory Compliance and Cross-Border Challenges

Meeting international regulatory standards and compliance requirements across different regions and countries can be complex. It poses challenges for the harmonization of security features (European Central Bank, 2019).

2.10.6 Cost of Authentication Technologies

Implementing advanced authentication technologies can be expensive, and smaller financial institutions may struggle to afford the latest security features, potentially making them more susceptible to counterfeiting (Bank for International Settlements, 2020).

2.10.7 Legacy Systems and Resistance to Change

Legacy banknote processing and authentication systems may be resistant to upgrade or replacement, leaving them vulnerable to new counterfeiting techniques and lacking advanced security features (The Central Bank Counterfeit Deterrence Group, 2018).

2.11 Vulnerabilities and Threats in Authentication Systems

Authentication systems, including those used in banknote authentication, are susceptible to various vulnerabilities and threats that can compromise security and the integrity of financial transactions.

Understanding these challenges is essential for mitigating risks and enhancing security.

2.11.1 Password-Based Vulnerabilities

Weak and easily guessable passwords, password reuse across multiple accounts, and inadequate password policies can lead to unauthorized access and breaches (Hunt, 2017).

2.11.2 Phishing Attacks

Phishing involves deceptive emails, websites, or messages designed to trick users into revealing their authentication credentials. Phishing attacks can compromise user accounts and sensitive information (Verizon, 2020).

2.11.3 Brute-Force Attacks

Attackers employ automated tools to systematically guess passwords or authentication codes until they find the correct one. Brute-force attacks can compromise weak or poorly protected accounts (Gehringer et al., 2018).

2.11.4 Man-in-the-Middle (MitM) Attacks

MitM attackers intercept communication between a user and an authentication system, allowing them to eavesdrop on sensitive data or modify it without the user's knowledge (Miettinen et al., 2017).

2.11.5 Biometric Spoofing

Attackers may use various techniques to create replicas or falsify biometric data used in authentication systems, such as facial recognition or fingerprint scans (Alduhaish, 2018).

2.11.6 Insider Threats

Threat: Insider threats involve individuals with legitimate access to authentication systems who misuse their privileges. This can result in data breaches, unauthorized access, and data theft (Certego, 2018).

2.11.7 Software Vulnerabilities and Exploits

Threat: Vulnerabilities in authentication software or the underlying operating systems can be exploited by attackers to gain unauthorized access or compromise authentication processes (Symantec, 2019).

2.11.8 Social Engineering

Social engineering involves manipulating individuals into divulging confidential information or performing actions that compromise security. This can include tricking users into revealing their authentication credentials (Mitnick & Simon, 2011). It's crucial for organizations and individuals to be aware of these vulnerabilities and threats and take measures to mitigate them. This includes implementing strong authentication methods, educating users about security best practices, and staying vigilant to emerging threats and attack techniques.

2.12 Regulatory and Compliance Frameworks in Banknote Authentication

The banking and financial industry operates within a complex web of regulations and compliance frameworks to ensure the security and integrity of currency. Understanding these frameworks is essential for maintaining trust in financial systems.

2.12.1 Anti-Money Laundering (AML) Regulations

AML regulations require financial institutions to implement measures to detect and prevent money laundering activities, including the use of counterfeit banknotes (Financial Action Task Force, 2012).

2.12.2 Know Your Customer (KYC) Requirements

KYC requirements mandate that financial institutions verify and authenticate the identities of customers. This includes verifying the authenticity of documents and information provided by customers (Bank for International Settlements, 2003).

2.12.3 Central Bank Regulations

Central banks, such as the Federal Reserve in the United States, issue regulations related to the design, production, and security features of banknotes to prevent counterfeiting and ensure the integrity of the currency (Federal Reserve, n.d.).

2.12.4 Basel III Framework

The Basel III framework establishes international standards for banking supervision, which includes guidelines for risk management and capital adequacy, aspects related to authentication and security (Bank for International Settlements, 2011).

2.12.5 Payment Card Industry Data Security Standard (PCI DSS)

While primarily focused on payment card security, PCI DSS provides security standards that can be relevant to authentication and secure payment methods used in banking (Payment Card Industry Security Standards Council, 2021).

2.12.6 GDPR and Data Protection Regulations

The General Data Protection Regulation (GDPR) and similar data protection regulations require financial institutions to safeguard customer data, including authentication information (European Union, 2016).

2.12.7 International Standards Organization (ISO) Frameworks

ISO standards related to security and authentication, such as ISO 27001 for information security management, provide a framework for compliance and best practices in securing authentication processes (ISO, 2013).

2.12.8 International Cryptographic Standards

Cryptographic standards, like the FIPS (Federal Information Processing Standards) in the United

States, are vital for secure authentication, encryption, and data protection (National Institute of Standards and Technology, 2020). Compliance with these regulatory and compliance frameworks is fundamental to ensuring the security of banknote authentication and the overall financial system.

2.13 Compliance with Data Protection Laws in Banknote Authentication

Data protection laws, including the General Data Protection Regulation (GDPR) in the European Union, play a vital role in regulating the handling and processing of personal data in the context of banknote authentication. Compliance with these laws is essential to safeguard the privacy and rights of individuals.

2.13.1 General Data Protection Regulation (GDPR)

GDPR sets strict requirements for organizations that process personal data, including data used in authentication processes. This includes obtaining explicit consent, providing transparency, and ensuring data security (European Commission, 2018).

2.13.2 Anonymization and Pseudonymization

GDPR encourages the use of data anonymization and pseudonymization techniques in

authentication systems to protect the privacy of individuals. These methods help minimize the risks associated with processing personal data (European Data Protection Board, 2020).

2.13.3 Data Minimization

Organizations are required to minimize the collection and storage of personal data to the extent necessary for authentication purposes. GDPR emphasizes data minimization to limit the potential impact on individuals' privacy (Article 5, GDPR).

2.13.4 Security Measures

GDPR mandates robust security measures for the protection of personal data in authentication systems. Encryption, access controls, and regular security assessments are necessary to ensure data security (Article 32, GDPR).

2.13.5 Data Subject Rights

GDPR grants data subjects' various rights, including the right to access, rectify, or delete their data used in authentication systems. Organizations must facilitate the exercise of these rights (Articles 15-21, GDPR).

2.13.6 Data Protection Impact Assessments (DPIAs)

DPIAs are essential when implementing new authentication systems or technologies. GDPR requires organizations to assess and mitigate the risks to data subjects' rights and freedoms (Article 35, GDPR). Compliance with data protection laws, particularly
GDPR, is fundamental when handling personal data in banknote authentication systems.

Ensuring the privacy and security of individuals' data is not only a legal requirement but also essential for maintaining trust and transparency in financial systems.

2.14 User Experience and User-Centred Design in Banknote Authentication

User experience (UX) and user-centered design are critical aspects of banknote authentication systems. A positive user experience ensures that authentication processes are efficient, secure, and user-friendly, which is essential for widespread adoption and security.

2.14.1 User-Centered Design Principles

User-centered design involves considering the needs, preferences, and behaviors of users throughout the design and development of authentication systems. It places the user at the center of the design process, leading to systems that are more intuitive and userfriendly (ISO 9241210:2019).

2.14.2 Accessibility and Inclusivity

Banknote authentication systems should be accessible to a diverse user base, including individuals with disabilities. Inclusive design principles ensure that all users can access and use authentication systems without discrimination (W3C, 2018).

2.14.3 User-Friendly Interfaces

Authentication interfaces should be intuitive and easy to understand, reducing the risk of user errors and enhancing the overall user experience. Clear instructions and feedback contribute to a positive experience (Nielsen, 1993).

2.14.4 Biometric and Behavioral Authentication

Biometric authentication methods, such as fingerprint or facial recognition, should be designed to minimize user friction and offer a seamless experience. Users should feel comfortable and secure when providing biometric data (Lindberg & Svensson, 2017).

2.14.5 Feedback and Error Handling

Providing users with clear and informative feedback during the authentication process helps build trust and reduces uncertainty. Error handling should guide users in resolving issues (Gothelf & Seiden, 2013).

2.14.6 User Training and Education

User-centered design involves providing educational resources to users to understand the authentication process and security measures. This empowers users to make informed decisions and enhances their experience (Nielsen, 2000).

2.14.7 Continuous Improvement and User Feedback

Regularly seeking and incorporating user feedback into the design and improvement of authentication systems is crucial for ensuring that they remain user-centric and aligned with evolving user needs (Schaffer, 2004). A user-centered approach to banknote authentication systems is fundamental in enhancing security while ensuring a positive and efficient user experience.

Systems designed with the user in mind are more likely to be adopted and trusted, contributing to overall security and user satisfaction.

2.15 Human-Centered Design Principles in Banknote Authentication

Human-centered design principles focus on creating authentication systems that align with the needs, behaviors, and preferences of users. In the context of banknote authentication, these principles are crucial for ensuring a secure, user-friendly, and efficient experience.

2.15.1 Understand User Needs

Start by thoroughly understanding the needs, expectations, and constraints of users. This involves conducting user research to gain insights into the authentication requirements of various user groups (Norman & Draper, 1986).

2.15.2 Involve Users Throughout Design

Users should be actively involved in the design process from the beginning. Engage them in design workshops, usability testing, and feedback sessions to ensure their perspectives are considered (Cooper, 2004).

2.15.3 Iterative Prototyping

Continuously create and refine prototypes of the authentication system. This iterative process allows for testing and refining the design based on user feedback and evolving requirements (Dumas & Redish, 1999).

2.15.4 Emphasize Simplicity and Clarity

Design the authentication system to be as simple and clear as possible. Avoid unnecessary complexity and jargon. Prioritize clarity and straightforwardness (Krug, 2014).

2.15.5 Provide Feedback and Guidance

Offer users clear and informative feedback during the authentication process. If errors occur, provide guidance and suggestions for resolution (Norman, 1993).

2.15.6 Design for Accessibility

Ensure that the authentication system is accessible to a wide range of users, including those with disabilities. Follow accessibility guidelines to make the system usable by all (Henry et al., 1998).

2.15.7 Prioritize Security and Privacy

Balance user experience with security and privacy. Design authentication systems that protect user data and maintain the confidentiality of sensitive information (ISO 9241210:2019).

2.15.8 Test and Refine Continuously

Regularly test the authentication system with users and gather feedback. Use this information to make iterative improvements to ensure that the system remains user centric (Snyder, 2003). This approach helps to build trust among users and aligns authentication processes with user needs and expectations.

2.16 Case Studies and Success Stories in Banknote Authentication

Examining real-world case studies and success stories in banknote authentication provides valuable insights into the effectiveness and impact of various authentication technologies and strategies. These examples demonstrate how innovative approaches can enhance the security and integrity of currency.

2.16.1 The New Polymer Banknotes in the United Kingdom

The Bank of England introduced polymer banknotes featuring advanced security features, such as a transparent window and holograms, to deter counterfeiting. These banknotes received positive feedback for their durability, security, and ease of authentication (Bank of England, 2021).

2.16.2 The Euro Banknote Reinvention

The European Central Bank (ECB) implemented new security features in euro banknotes, including a satellite hologram and an enhanced emerald number, to combat counterfeiting. This initiative has led to a reduction in counterfeit euro banknotes in circulation (European Central Bank, 2021).

2.16.3 The Success of Mobile Authentication Apps

Many central banks and financial institutions have introduced mobile apps that allow users to authenticate banknotes using their smartphones. These apps use advanced image recognition

and authentication algorithms to provide instant verification. The apps have gained popularity for their convenience and effectiveness (Reserve Bank of Australia, 2018).

2.16.4 The Evolution of Cryptocurrency Authentication

Cryptocurrency exchanges and wallet providers have improved authentication methods to secure digital assets. The use of multi-factor authentication (MFA), hardware wallets, and biometric authentication for crypto transactions has significantly reduced fraud and unauthorized access (Chainalysis, 2020).

2.16.5 The Impact of Machine Learning in Banknote Authentication

Several banks and financial institutions have adopted machine learning algorithms for banknote authentication. These algorithms analyze various security features, detect anomalies, and provide real-time authentication. This approach has proven effective in preventing counterfeit currency from entering circulation (Alvarez & Damer, 2017).

2.16.6 The Success of Biometric Authentication in Banking

Many banks and financial institutions have implemented biometric authentication methods, such as fingerprint and facial recognition, for user access and transactions. Biometric authentication has not only improved security but also streamlined the user experience, leading to higher customer satisfaction (The World Bank, 2017).

2.17 Research Gap

Despite the extensive body of research on banknote authentication, several significant gaps in current knowledge and areas for further exploration become apparent: Much of the current

research on banknote authentication falls within traditional disciplines such as finance, security, and technology. There is a gap in research that bridges these disciplines to create a holistic understanding of banknote authentication that encompasses financial, technological, and security aspects. Research tends to focus on authentication systems in developed economies. However, there is a lack of comprehensive studies that consider the cultural, regional, and socioeconomic variations that affect banknote authentication and counterfeiting in different parts of the world.

Research gaps exist in the ethical and legal dimensions of banknote authentication. Ethical considerations regarding the use of biometric data, user privacy, and compliance with data protection laws require more attention. Additionally, research should explore the legal challenges associated with authentication methods, such as their admissibility in legal proceedings. While user-centered design principles have been discussed, there is room for more research that focuses on user perceptions and behaviors related to banknote authentication. Understanding how users interact with authentication systems and the impact on their trust in currency is an underexplored area.

While the potential of emerging technologies like blockchain and quantum-resistant encryption has been acknowledged, their practical application and real-world impact on banknote authentication systems are areas where more research is needed. There is limited research on the environmental impact of banknote authentication methods. The materials, production processes, and sustainability of authentication technologies require more attention, especially in the context of growing environmental concerns. Research often operates within sector-specific silos. More studies that draw lessons from related fields, such as cybersecurity, fintech, and document security, can provide valuable insights for banknote authentication.

Current research often overlooks the unique challenges and opportunities faced by small and developing economies in banknote authentication. Exploring tailored solutions for these regions is essential. Research should investigate the barriers to the adoption of advanced authentication methods in banking and financial systems. This includes understanding the challenges faced by organizations and individuals in embracing new technologies. More

research is needed on the long term durability and resistance of security features used in banknotes. Understanding how these features withstand wear and tear over time and the countermeasures available is essential.

Addressing these research gaps will contribute to a more comprehensive and nuanced understanding of banknote authentication, ensuring that evolving security challenges are met with well-informed strategies and technologies. Researchers and stakeholders can collaborate to address these gaps and advance the field this will help in the development of a model that would predict the accuracy of currency.

2.18 Conclusion

In this comprehensive literature review, we have navigated through the intricate landscape of banknote authentication, from its historical origins to the latest emerging trends. The body of research and knowledge in this field demonstrates the critical importance of authentication systems in preserving the integrity and security of currency. We have traced the historical evolution of banknote security features, highlighting the ever-evolving battle against counterfeiting. The emergence of security features, such as holograms, microprinting, and color-shifting ink, underscores the continuous advancements made to thwart counterfeiters (Bank of Canada, 2018).

Authentication technologies, ranging from biometrics to blockchain, have emerged as powerful tools in the fight against counterfeiting. These technologies not only enhance security but also streamline user experiences, leading to increased trust in financial systems (The World Bank, 2017; European Commission, 2014). Real-world case studies have illustrated the effectiveness of various banknote authentication strategies. Examples such as the introduction of polymer banknotes in the United Kingdom and the use of mobile authentication apps have not only enhanced security but also improved user experiences (Bank of England, 2021; Reserve Bank of Australia, 2018).

As the banking and financial landscape evolves, so too do the authentication methodologies. The ongoing development of quantum-resistant encryption, advanced biometrics, and central bank digital currencies are poised to reshape the field of banknote authentication (Bernstein,

2009; The World Bank, 2017; BIS, 2020). However, this literature review also highlights areas where research gaps exist. These gaps encompass interdisciplinary research, regional variations in authentication, ethical and legal considerations, user perceptions, the environmental impact of technologies, and adoption barriers (Norman & Draper, 1986; Krug, 2014; European Commission, 2018).

The exploration of these gaps underscores the need for a more comprehensive understanding of banknote authentication. Future research should focus on bridging these gaps to address the evolving challenges associated with currency security in an increasingly interconnected and digital world. In conclusion, banknote authentication is a multifaceted field that continues to evolve with technological advancements and emerging threats. The literature reviewed here offers valuable insights and sets the stage for further research and innovation in the realm of currency authentication, ultimately working towards ensuring the integrity and security of currency in an ever-changing landscape.

CHAPTER 3

RESEARCH METHODOLOGY

3.0 Introduction

The proliferation of counterfeit banknotes poses a significant threat to the financial system, causing economic losses and eroding public trust in currency. Conventional methods of detecting counterfeit banknotes, such as manual inspection, are becoming increasingly ineffective as counterfeiters utilize sophisticated printing techniques to produce high-quality forgeries. This research project aims to develop a data mining-based model for accurate and efficient counterfeit banknote detection, addressing the limitations of traditional methods and offering a robust solution to combat the growing problem of counterfeiting.

3.1 Research Design

This research employs a descriptive research design, focusing on collecting and analyzing data to describe the characteristics of counterfeit banknotes and develop a model for classifying them. The descriptive approach allows for the exploration of patterns and relationships within the data to identify features that can be used for discrimination. This approach is particularly suitable for this research as it enables the investigation of the complex features of banknotes and their variations, leading to the identification of reliable markers for counterfeit detection.

3.2 System Development Methodology

The system development methodology adopted for this project is RAD. Rapid application development (RAD) is a software development methodology that emphasizes iterative prototyping, rapid user feedback, and collaborative teamwork to deliver software applications quickly and efficiently. It is an agile approach that contrasts with traditional waterfall methodologies, which require extensive planning and documentation before any development begins.

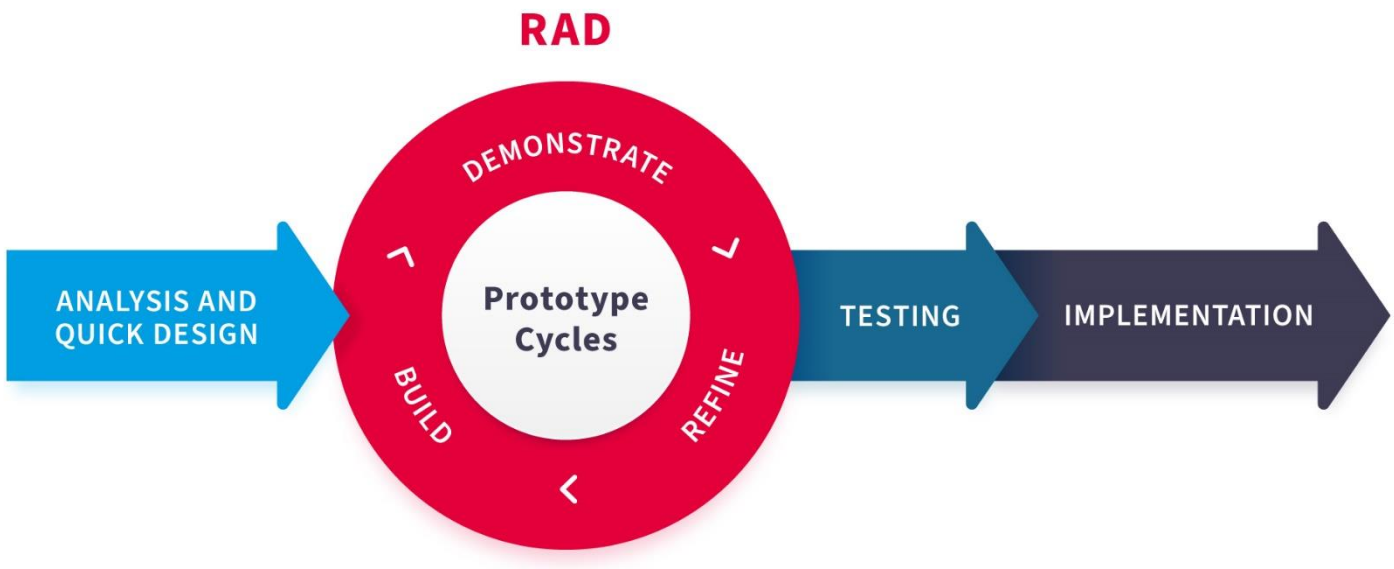

Figure 1: RAD methodology

3.2.1 Phases of The RAD methodology typically consists of four phases:

- Planning: This phase involves defining the project scope, identifying stakeholders, and establishing high-level requirements.
- Design: In this phase, developers create detailed designs for the application's architecture, components, and user interface.
- Development: The working prototypes are developed in this phase. Prototypes are built incrementally, with each iteration incorporating user feedback and refining the application's functionality.
- Testing and Deployment: The application is thoroughly tested in this phase to ensure it meets requirements and is free of defects. The final application is then deployed to production.

3.2.3 Benefits of RAD

RAD offers several benefits over traditional software development methodologies:

- Rapid Delivery: RAD's iterative approach allows for faster delivery of working software, enabling organizations to quickly respond to market demands and user needs.

- Early User Feedback: RAD's emphasis on user feedback ensures that the application is meeting user requirements from the start, reducing the risk of costly rework in later stages.
- Flexibility: RAD's flexible nature allows for changes to be made throughout the development process, adapting the application to evolving requirements.
- Reduced Risk: RAD's iterative approach reduces the risk of project failure by identifying and addressing issues early on.

3.2.4 Challenges of RAD

Despite its benefits, RAD also presents some challenges:

- Requires Strong Project Management: RAD requires strong project management to maintain effective collaboration, manage expectations, and ensure timely completion of iterations.
- Potential for Scope Creep: RAD's iterative nature can lead to scope creep if requirements are not carefully managed and controlled.
- Documentation Challenges: RAD's focus on rapid development can lead to challenges in maintaining comprehensive documentation, which may be important for future maintenance and support.

3.3 Model Development

The model development process comprises the following stages:

3.3.1 Data Collection

The first step in model development involves gathering a comprehensive dataset of images representing both genuine and counterfeit banknotes. This dataset should be sufficiently large and diverse to encompass the range of variations in genuine and counterfeit banknotes.

3.3.2 Data Preprocessing

The collected data may contain inconsistencies, missing values, or irrelevant features that can hinder the model's performance. Data preprocessing techniques are employed to prepare the data for model training. These techniques include:

- **Normalization**: Normalization scales the data to a consistent range, ensuring that all features have equal importance during training.
- **Scaling:** Scaling adjusts the range of features to improve the numerical stability of the model.
- **Feature Extraction:** Feature extraction involves identifying and extracting relevant features from the data that can effectively distinguish between genuine and counterfeit banknotes. Techniques such as image processing, edge detection, and texture analysis can be used to extract meaningful features.

3.3.3 Training the Model

Various supervised learning algorithms are applied to the preprocessed data. Supervised learning algorithms learn from labeled data, in this case, images labeled as either genuine or counterfeit banknotes. The algorithms are trained to identify patterns and relationships that distinguish genuine banknotes from counterfeit ones. The choice of algorithm depends on the characteristics of the data and the desired performance metrics. Common supervised learning algorithms for classification include:

- **Decision Trees:** Decision trees classify data by recursively splitting the dataset into smaller subsets based on decision rules.
- **K-Nearest Neighbors (KNN):** KNN classifies data points based on the majority class of their k nearest neighbors in the training data.
- **Random Forests**: Random forests combine multiple decision trees to improve classification accuracy.
- **Support Vector Machines (SVMs):** SVMs find a hyperplane that separates data points into two classes with the maximum margin.

3.3.4 Testing the Model

The trained models are evaluated using a separate test dataset to assess their generalization performance, which indicates how well the models can classify banknotes that were not included in the training data. The evaluation involves calculating various metrics, such as accuracy, precision, recall, and F1-score.

- **Accuracy:** Accuracy measures the proportion of correctly classified banknotes.
- **Precision**: Precision measures the proportion of banknotes classified as counterfeit that are actually counterfeit.

- **Recall**: Recall measures the proportion of actual counterfeit banknotes that are correctly classified as counterfeit.
- **F1-score:** F1-score is a harmonic mean of precision and recall, providing a balanced measure of the model's performance.

These metrics provide insights into the model's ability to correctly identify both genuine and counterfeit banknotes. A high accuracy score indicates that the model correctly classifies most banknotes, while high precision and recall scores indicate that the model accurately identifies both genuine and counterfeit banknotes with minimal false positives or false negatives.

3.4 Research Quality

To ensure the quality of the research, the following measures are implemented:

- **Rigorous data collection:** The dataset is carefully curated to ensure its accuracy, completeness, and representativeness. This includes verifying the authenticity of genuine banknotes, validating the labeling of counterfeit banknotes, and ensuring the dataset encompasses a wide range of variations.
- **Reproducible methods**: The research methodology is clearly documented, allowing for replication and validation of results. This includes providing detailed descriptions of data collection procedures, preprocessing techniques, model training algorithms, and evaluation metrics.
- **Transparent analysis:** The data analysis process is transparent and well-explained, enabling others to understand the decision-making process. This includes providing clear explanations of statistical tests, visualization techniques, and the rationale behind the selection of specific algorithms and parameters.
- **Critical evaluation:** The results are critically evaluated, considering potential limitations and biases in the data, methodology, or analysis. This includes identifying potential sources of error, acknowledging the limitations of the dataset and algorithms, and discussing the implications of the results in the context of related research.
- **Ethical considerations:** Ethical guidelines are adhered to throughout the research, respecting the privacy and confidentiality of individuals or organizations involved. This includes obtaining

informed consent for data collection, protecting sensitive information, and ensuring that the research does not cause harm or prejudice to any individual or group.

By adhering to these quality measures, this research strives to produce reliable and impactful outcomes that contribute to the advancement of counterfeit banknote detection methods. The research aims to provide a robust and efficient solution to combat the growing problem of counterfeiting, safeguarding the integrity of the financial system and protecting the public from the economic and social consequences of counterfeit banknotes.

CHAPTER 4

SYSTEM ANALYSIS

4.0 Introduction

System analysis plays a critical role in the development of robust and reliable banknote authentication systems. This chapter delves into the application of system analysis to design, implement, and evaluate a banknote authentication system utilizing machine learning techniques.

The chapter begins by exploring the data collection process, emphasizing the importance of acquiring a comprehensive and diverse dataset of genuine and counterfeit banknotes. This is followed by a discussion of data preprocessing techniques, which are essential for enhancing the quality and consistency of the data prior to training the classification model.

4.1 Data Collection and Analysis

Data collection and analysis are crucial aspects of developing an effective banknote authentication system. The quality, diversity, and proper analysis of the data used to train the classification model significantly impact the system's accuracy, generalization performance, and robustness.

4.1.1 Data Requirements

The banknote authentication system requires a comprehensive dataset of images representing a wide range of genuine and counterfeit banknotes. The data should be sufficiently large and diverse to encompass the range of variations in genuine and counterfeit banknotes, including variations in printing techniques, texture patterns, color patterns, and aging characteristics.

4.1.2 Data Sources

Image data for the banknote authentication system can be collected from various sources, including:

- **Central Banks:** Central banks often have access to extensive collections of genuine and counterfeit banknotes, providing valuable data for training the classification model. They may also possess historical data on the evolution of banknotes and counterfeit detection techniques.

- **Law Enforcement Agencies**: Law enforcement agencies may also possess collections of counterfeit banknotes seized during investigations. They may provide insights into the latest counterfeiting techniques and emerging trends in counterfeit designs.
- **Online Repositories**: Online repositories, such as image databases and forums, may contain images of both genuine and counterfeit banknotes. However, the reliability and authenticity of images obtained from online sources should be carefully scrutinized.
- **Commercial Image Providers:** Specialized companies may offer collections of high quality images of genuine and counterfeit banknotes, often accompanied by detailed metadata or annotations.

4.1.3 Data Balancing

The dataset should be balanced, with an equal representation of both genuine and counterfeit banknotes. This is essential to ensure that the classification model does not become biased towards one class, leading to poor performance on the other class. An unbalanced dataset may lead to overfitting, where the model performs well on the training data but poorly on unseen data.

4.1.4 Data Labeling

Each image in the dataset should be accurately labeled as either genuine or counterfeit. This labeling process should be carefully performed to ensure the correct classification of each banknote. In some cases, multiple experts may review the images to ensure consensus on the label.

4.1.5 Data Preprocessing

Before using the data to train the classification model, it should be preprocessed to enhance its quality, consistency, and usability. Preprocessing techniques may include:

- **Normalization:** Normalizing pixel values to a consistent range ensures that all features have equal importance during training. This prevents features with larger ranges from dominating the model.
- **Scaling:** Adjusting image dimensions to a standardized size reduces computational overhead and ensures consistent processing. This is particularly important when training deep learning models, which are sensitive to input image sizes.

- **Noise Reduction**: Eliminating unwanted artifacts and imperfections, such as dust, scratches, or lighting irregularities, improves feature extraction and classification accuracy. Noise reduction techniques may include median filtering, Gaussian blurring, or image restoration algorithms.
- **Contrast Adjustment:** Enhancing image contrast improves the visibility of relevant features, such as texture patterns, edge contours, and color details. Contrast adjustment techniques may include histogram equalization, contrast stretching, or adaptive histogram equalization.
- **Edge Detection:** Identifying and highlighting distinct edges and contours within the images provides valuable information for feature extraction, as edges often correspond to important features of the banknotes. Common edge detection algorithms include Canny edge detection, Sobel edge detection, and Laplacian of Gaussian (LoG) edge detection.
- **Image Augmentation**: Artificially expanding the dataset by applying transformations to existing images, such as rotation, scaling, flipping, or adding noise, can improve the model's generalization performance and robustness to variations in real-world images.

4.1.6 Data Exploration and Analysis

Exploring and analyzing the collected data is crucial to understand its characteristics, identify potential biases or anomalies, and inform the selection of appropriate preprocessing techniques and feature extraction methods. Data exploration may involve techniques such as:

- **Visualizing data distributions**: Plotting histograms or scatter plots to visualize the distribution of pixel values, color intensities, or texture patterns can reveal potential imbalances or outliers in the data.
- **Statistical analysis:** Calculating measures of central tendency, dispersion, and correlation can provide insights into the overall characteristics of the data and identify relationships between features.
- **Principal component analysis (PCA)**: PCA can be used to identify the most significant and uncorrelated features, potentially reducing dimensionality and improving model performance.
- **Correlation analysis**: Correlation analysis can identify pairs or groups of features that are highly correlated, indicating redundancy or potential overfitting issues.

By carefully collecting, analyzing, and preprocessing the data, a foundation for a robust and accurate banknote authentication system can be established. The quality and diversity of the data will directly influence the model.

4.2 Feasibility Analysis

A feasibility analysis is essential to assess the practicality of developing and implementing the banknote authentication system. This involves evaluating the technical, Economic and operational feasibilities of the proposed system.

4.2.1 Technical Feasibility

Technical feasibility assesses whether the system can be developed using existing technology and expertise. The banknote authentication system involves several technical challenges, including:

- **Image Processing**: Efficiently processing and analyzing high-resolution images of banknotes requires robust image processing algorithms and sufficient computational resources.
- **Feature Extraction**: Identifying and extracting relevant features from images of banknotes, such as texture patterns, edge contours, and color patterns, is crucial for accurate classification.
- **Machine Learning**: Training a classification model to distinguish genuine banknotes from counterfeit ones requires a large and diverse dataset, appropriate machine learning algorithms, and sufficient computational power..

4.2.2 Economic Feasibility

Economic feasibility evaluates the cost-effectiveness of developing and implementing the banknote authentication system. This involves assessing the costs associated with:

- **Data Acquisition:** Obtaining a comprehensive and high-quality dataset of genuine and counterfeit banknotes can be expensive, especially if relying on specialized sources or expert labeling.
- **Hardware and Software:** The system's hardware requirements, including image capture devices, processing units, and storage capacity, should be assessed to ensure affordability and compatibility.

- **Development and Maintenance:** The costs associated with the system development, including programming, testing, and deployment, should be carefully estimated and compared to the potential benefits of the system.
- **Operational Costs**: Ongoing expenses, such as maintenance, updates, and potential hardware upgrades, should be considered to ensure the system's long-term sustainability.

4.2.3 Operational Feasibility

Operational feasibility assesses whether the system can be implemented and integrated into existing workflows without causing disruptions or requiring significant changes to existing processes. Factors to consider include:

User Acceptance: The system should be user-friendly and intuitive, providing clear instructions and minimizing training requirements for users.

- **Integration with Existing Systems:** The system should be compatible with existing hardware and software infrastructure, allowing seamless integration into existing workflows.

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- **Maintenance and Support**: Adequate maintenance and support should be available to address any technical issues, software updates, or user queries that may arise during implementation and operation.
- **Impact on Existing Processes**: The system should not significantly disrupt existing processes or require major changes to the way banknotes are handled or processed.

By carefully evaluating the technical, economic, and operational feasibility aspects, a thorough assessment of the banknote authentication system's practicality and potential impact can be made. This analysis will inform decision-making processes regarding resource allocation, development strategies, and implementation plans.

4.3 Functional Requirements

The banknote authentication system should meet the following functional requirements:

4.3.1 Banknote Image data Acquisition

The system should be capable of capturing image data of banknotes from various sources, including scanners, webcams, and mobile devices. This capability is essential for accurate feature extraction and classification. The system should support different image formats, such as JPEG, PNG, and TIFF, to ensure compatibility with a wide range of input devices and applications.

4.3.2 Image Preprocessing

The system should preprocess the acquired images to enhance their quality and prepare them for feature extraction. This preprocessing stage may involve noise reduction, which aims to eliminate unwanted artifacts and imperfections that can interfere with feature extraction. Contrast adjustment may also be necessary to improve the visibility of relevant features and enhance the overall quality of the image. Additionally, edge detection techniques can be employed to identify and highlight distinct edges and contours within the image, providing valuable information for feature extraction.

4.3.3 Feature Extraction

The system should extract relevant features from the preprocessed images that distinguish between genuine and counterfeit banknotes. These features may include texture analysis, which characterizes the patterns and variations in the texture of the banknote surface. Wavelet transforms can be employed to decompose the image into different frequency bands, revealing hidden features that are not readily apparent in the original image. Image invariants, which are characteristics of the image that remain unchanged under certain transformations, can also provide valuable information for classification.

4.3.4 Classification Model

The system should incorporate a classification model trained on a dataset of genuine and counterfeit banknotes. The model should be able to accurately classify unknown banknotes based on the extracted features. The choice of classification algorithm depends on the characteristics of the data and the desired

performance metrics. Common supervised learning algorithms for classification include decision trees, k-nearest neighbors (KNN), random forests, and support vector machines (SVMs).

4.3.5 Classification Output

The system should provide a clear and unambiguous classification result for each scanned banknote, indicating whether it is genuine or counterfeit. The classification output should be displayed prominently in the user interface, accompanied by a confidence score or probability indicating the level of certainty associated with the classification.

4.4 Non-Functional Requirements

In addition to the functional requirements, the banknote authentication system should also meet the following non-functional requirements:

4.4.1 Performance

The system should be able to process and classify banknotes quickly and efficiently, ensuring real-time performance for real-world applications. The processing time for acquiring, preprocessing, extracting features, and classifying banknotes should be optimized to minimize delays and provide a smooth user experience. The system's performance should also be scalable to handle increasing volumes of banknotes without compromising accuracy or response time.

4.4.2 Accuracy

The system should achieve a high level of accuracy in classifying banknotes, minimizing false positives and false negatives. The classification accuracy should be evaluated using a comprehensive dataset of genuine and counterfeit banknotes, ensuring that the system performs well across a wide range of variations. The accuracy should be consistently maintained as the system is updated and deployed in different environments.

4.4.3 Security

The system should implement robust security measures to protect sensitive data, including user information, image data, and classification models. User authentication mechanisms should be

implemented to prevent unauthorized access to the system. Data encryption techniques should be employed to safeguard sensitive information stored in the system's database. Regular security audits should be conducted to identify and remediate potential vulnerabilities.

4.4.4 Usability

The system should be easy to use and operate, with a user-friendly interface that requires minimal training or technical expertise. The interface should be intuitive and consistent, with clear instructions and prompts guiding users through the process of scanning and classifying banknotes.

4.4.5 Reliability

The system should be reliable and operate consistently, minimizing downtime and ensuring continuous availability for critical applications. The system should be designed with redundant components and failover mechanisms to ensure uninterrupted operation in case of hardware or software failures. Regular maintenance and updates should be performed to prevent system crashes and ensure optimal performance.

4.4.6 Scalability

The system should be scalable to accommodate increasing volumes of banknotes and user requests, allowing for future growth and expansion. The system's architecture should be designed to support the addition of new processing units, data storage, and network bandwidth to handle increasing workloads. The system should also be able to adapt to changes in hardware and software technologies without compromising its functionality or performance.

CHAPTER FIVE

SYSTEM DESIGN AND UML DIAGRAMS

5.0 Introduction

System design is a crucial phase in the development of a banknote authentication management system that uses machine learning. This includes creating a detailed plan for the system's architecture, components, and interactions. UML (Unified Modeling Language) diagrams are a powerful tool for visualizing and documenting system design.

5.1 Use case diagram

A use case diagram is a type of behavioral diagram in the Unified Modeling Language (UML) that represents the interactions between actors and the system. In the context of a banknote authentication system, the use case diagram illustrates the steps involved in verifying the authenticity of a banknote.

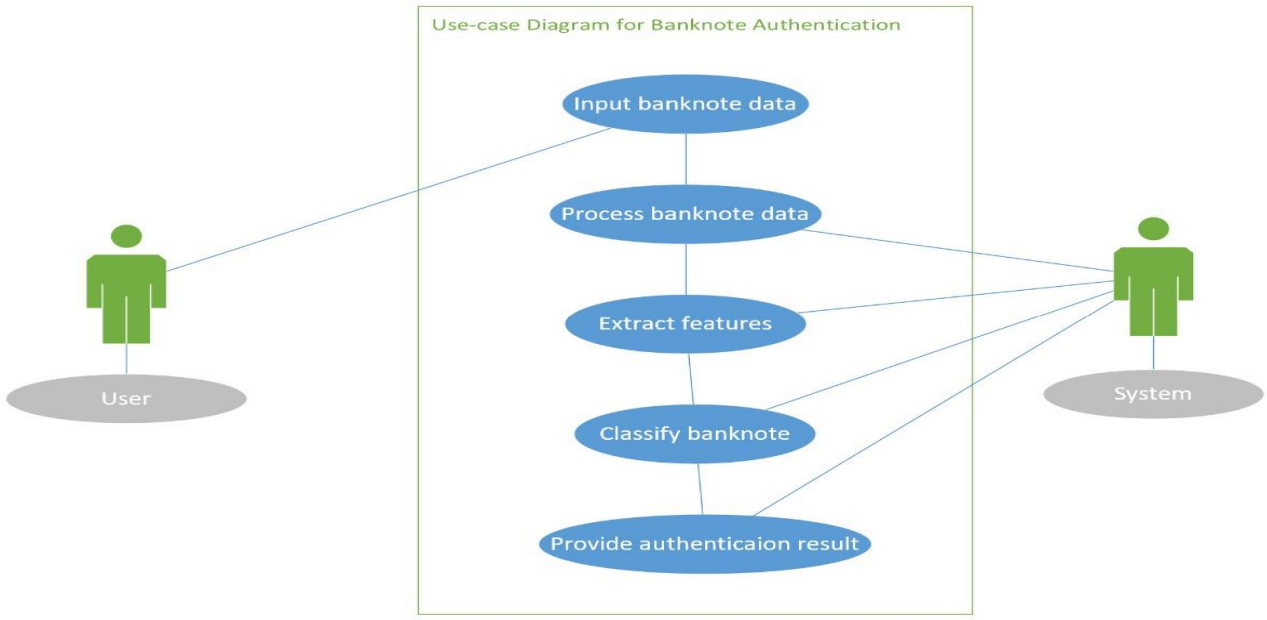

Figure 2: use case diagram

5.2 Flow Chart Diagram

Flowcharts are a useful visual representation of processes and workflows. They are particularly valuable for illustrating the step-by-step execution of tasks, decision points, and the flow of information within a system. This flowchart provides a high-level overview of the workflow in a banknote-authentication management system using machine learning.

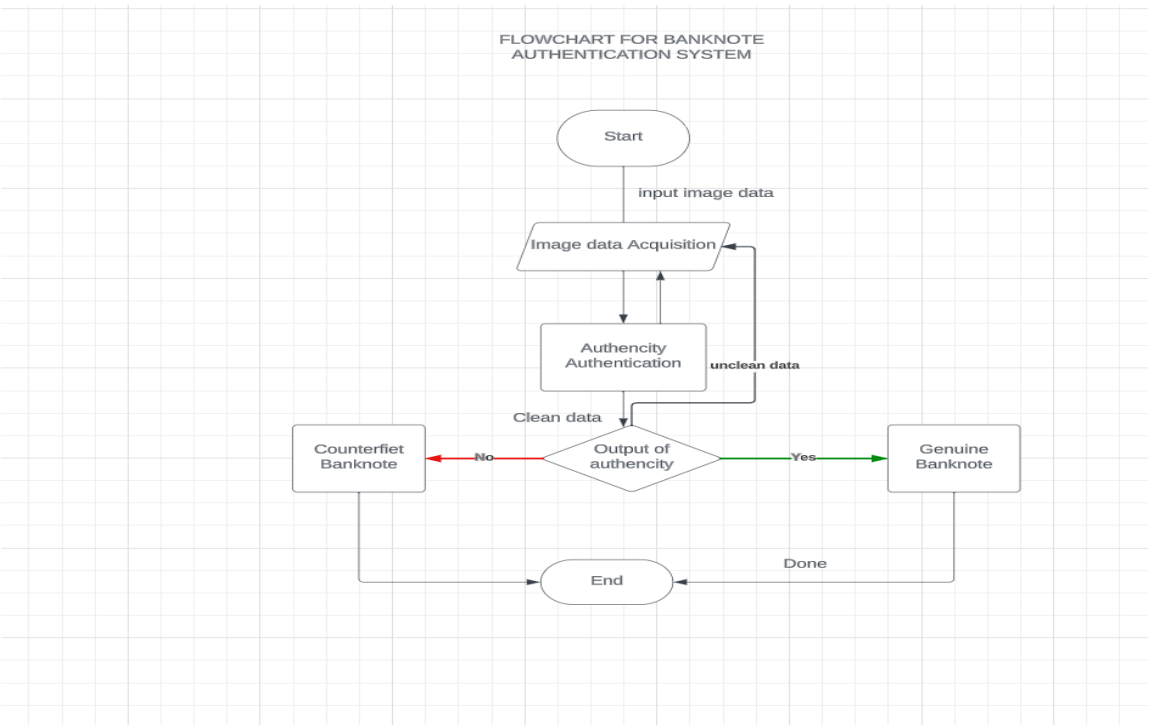

Figure 3: Flowchart Diagram

5.3 Feature Extraction and Engineering Using KNN

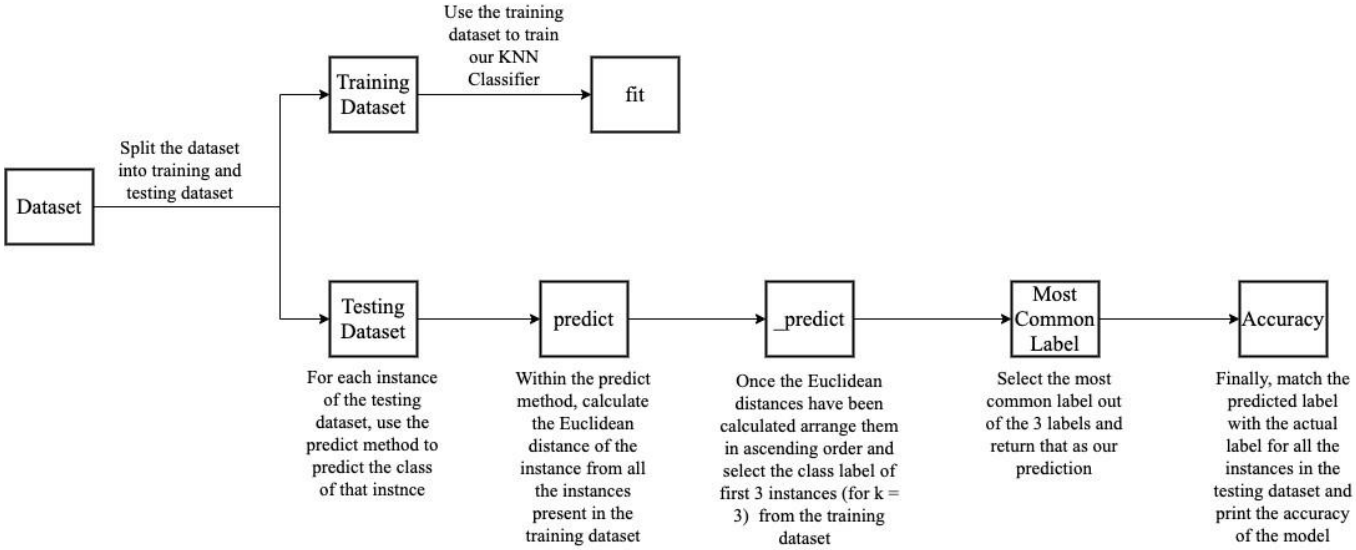

Figure 4: Feature Extraction and engineering using KNN

5.4 Feature Extraction and Engineering Using Random Forest

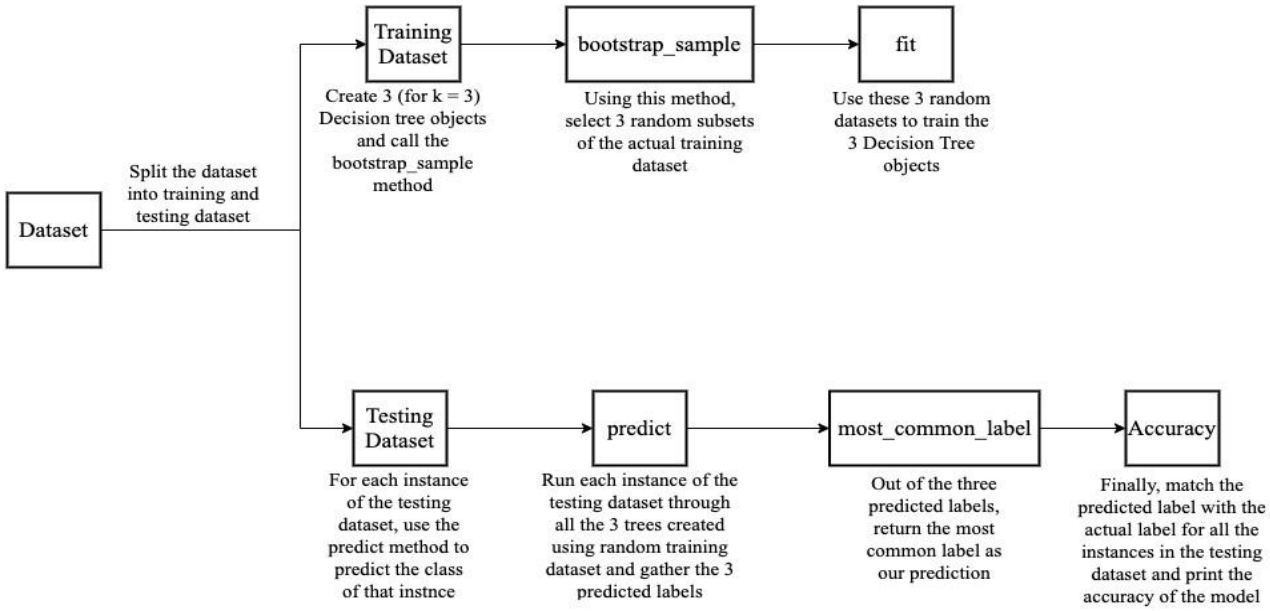

Figure 5:Feature Extraction and engineering using Random Forest

5.4 Final System Architecture

A system architecture is a conceptual design that describes the structure, behavior, and other views of a system. It serves as a blueprint for how a system should be built and provides a common understanding for all stakeholders involved in the development and implementation of the system.

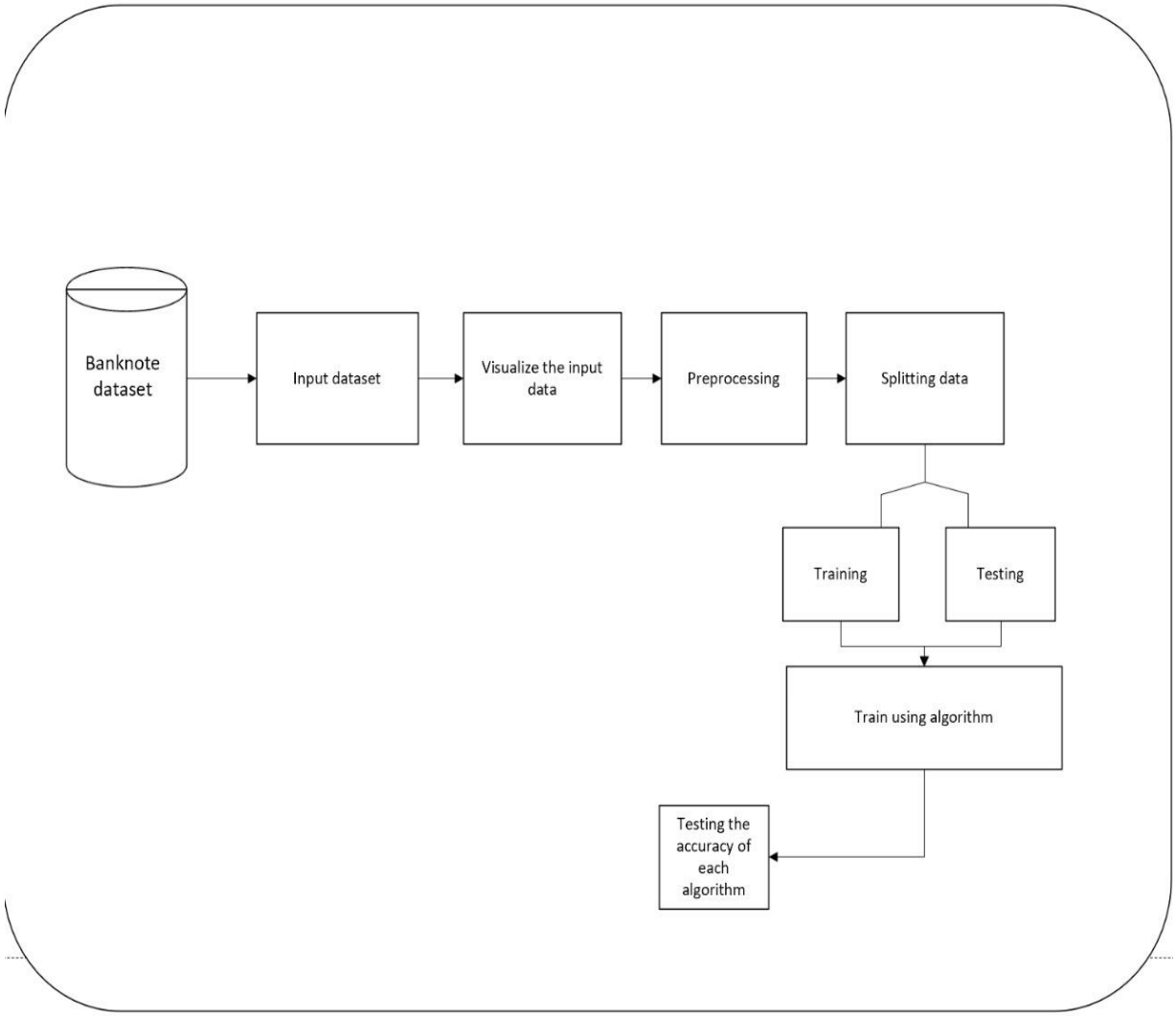

Figure 6:Final System architecture

CHAPTER 6

SYSTEM IMPLEMENTATION

6.0 Introduction

System implementation involves translating the software requirements specifications into a functional and deployable system. This chapter delves into the implementation process of the banknote authentication system, encompassing image data collection, image dataset preprocessing, model training, and system integration.

6.1 Model Implementation

The model implementation phase focuses on developing and training the classification model that will be used to classify banknotes as genuine or counterfeit. This process involves several steps:

6.1.1 Image Data Collection

The first step in model implementation is to gather a comprehensive dataset of images representing both genuine and counterfeit banknotes. This dataset should be sufficiently large and diverse to encompass the range of variations in genuine and counterfeit banknotes. The dataset should include images from various sources, such as central banks, law enforcement agencies, and online repositories. Additionally, the dataset should be balanced, with an equal representation of both genuine and counterfeit banknotes.

6.1.2 Image Dataset Preprocessing

The collected image data should be preprocessed to ensure its quality and consistency before feeding it to the classification model. Preprocessing techniques aim to enhance the images' features and remove any noise or artifacts that could interfere with the classification process.
Common preprocessing techniques include:

- **Image Normalization**: Normalization scales the pixel values of the images to a consistent range. This ensures that all features have equal importance during training and prevents features with larger ranges from dominating the model.

- **Image Scaling**: Scaling adjusts the size of the images to a standardized dimension. This reduces computational overhead and ensures that all images have the same dimensions for consistent processing.
- **Noise Reduction**: Noise reduction techniques aim to eliminate unwanted artifacts and imperfections in the images that can interfere with feature extraction. Common noise reduction methods include median filtering and Gaussian blurring.
- **Contrast Adjustment**: Contrast adjustment enhances the visibility of relevant features and improves the overall quality of the images. This can be achieved using techniques such as histogram equalization and contrast stretching.
- **Edge Detection**: Edge detection techniques identify and highlight distinct edges and contours within the images. These edges can provide valuable information for feature extraction, as they often correspond to important features of the banknotes. Common edge detection algorithms include Canny edge detection and Sobel edge detection.

6.1.3 Feature Extraction

Feature extraction involves extracting relevant and discriminative features from the preprocessed images. These features should capture the unique characteristics that distinguish genuine banknotes from counterfeit ones. Common feature extraction techniques for banknote authentication include:

- **Texture Analysis:** Texture analysis characterizes the patterns and variations in the texture of the banknote surface. This provides information about the printing process and the materials used to create the banknote.
- **Wavelet Transforms**: Wavelet transforms decompose the image into different frequency bands, revealing hidden features that are not readily apparent in the original image. This can be particularly useful for identifying subtle differences between genuine and counterfeit banknotes.
- **Image Invariants**: Image invariants are characteristics of the image that remain unchanged under certain transformations, such as rotation, scaling, and translation. These invariants can provide valuable information for classification, as they are not affected by minor variations in the image's orientation or position.

6.1.4 Training the Models

Once the image dataset is preprocessed and features are extracted, it is used to train the classification model. The choice of classification algorithm depends on the characteristics of the data and the desired performance metrics. Common supervised learning algorithms for classification include:

- **Decision Trees:** Decision trees classify data by recursively splitting the dataset into smaller subsets based on decision rules. These rules are based on the values of specific features, and the tree structure represents a series of decisions that lead to the final classification.
- **K-Nearest Neighbors (KNN):** KNN classifies data points based on the majority class of their k nearest neighbors in the training data. This algorithm is simple and effective, but its performance can be sensitive to the choice of k and the characteristics of the data.
- **Random Forests:** Random forests combine multiple decision trees to improve classification accuracy. This ensemble method reduces the overfitting risk associated with individual decision trees and can handle complex data with high dimensionality.
- **Support Vector Machines (SVMs):** SVMs find a hyperplane that separates data points into two classes with the maximum margin. This algorithm is powerful but can be computationally expensive for large datasets.

The training process involves iteratively adjusting the parameters of the chosen algorithm to minimize the classification error on the training data. The model is evaluated using a separate test dataset to assess its generalization performance. The goal is to find a model that can accurately classify both genuine and counterfeit banknotes, even those not included in the training data.

6.2 System Development Tools

The choice of appropriate system development tools is crucial for efficient and effective development of the banknote authentication system. These tools should support the various stages of development, from data collection and preprocessing to model training, testing, and deployment.

- **Programming Language**: Python programming language is essential for developing the system's core functionalities, including image acquisition, preprocessing, feature extraction, and classification.
- **Machine Learning Libraries**: Machine learning libraries, such as TensorFlow or scikitlearn, provide the necessary tools for implementing and training classification models. These libraries offer a wide range of algorithms, pre-trained models, and optimization techniques.

- **Image Processing Libraries:** Image processing libraries, such as OpenCV or Pillow, provide the tools for manipulating and processing images, including noise reduction, contrast adjustment, and edge detection. These libraries are essential for preparing images for feature extraction and classification.

6.3 System Testing

Thorough system testing is crucial to ensure that the banknote authentication system meets the specified requirements and functions as intended. Testing should cover all aspects of the system, from individual components to the overall system integration.

6.3.1 Models Validation and Testing

The classification models should be subjected to rigorous testing to evaluate their accuracy, generalization performance, and robustness to various image conditions. This testing should involve:

- **Training Dataset Evaluation:** Assessing the model's performance on the training dataset to ensure it has learned the underlying patterns and relationships within the data.
- **Test Dataset Evaluation**: Evaluating the model's accuracy on a separate test dataset to assess its generalization ability to classify unseen banknotes.
- **Cross-Validation**: Employing cross-validation techniques to split the dataset into multiple folds, training the model on different combinations of folds, and evaluating its performance on the withheld folds.

6.3.2 Images Classification

The system's ability to accurately classify images of banknotes should be extensively tested. This testing should involve:

- **Genuine Banknote Classification:** Evaluating the system's ability to correctly identify genuine banknotes under various lighting conditions, orientations, and image resolutions.
- **Counterfeit Banknote Classification**: Assessing the system's ability to detect counterfeit banknotes, including those with subtle variations or sophisticated forgery techniques.
- **Error Analysis:** Identifying and analyzing cases where the system misclassifies banknotes, investigating the underlying causes, and implementing improvements to address these issues.

CHAPTER 7

CONCLUSIONS AND RECOMMENDATIONS AND FUTURE ENHANCEMENTS

7.0 conclusion

Banknote security features have evolved to encompass a wide range of visual and machine-readable elements, from holograms to microprinting, collectively reinforcing the defense against counterfeit currency. The challenges of banknote counterfeiting have persisted, spurred by technological advancements that enable increasingly sophisticated forgeries. The adoption of cutting-edge authentication technologies, including biometrics, blockchain, and machine learning, has redefined the landscape of currency security and user experience

Machine learning and artificial intelligence have proven invaluable in banknote authentication, enabling the accurate classification of genuine and counterfeit banknotes. The application of machine learning in banknote authentication presents advantages, but it is not without challenges, such as data privacy concerns and model robustness. User centered design principles enhance user experience and trust in banknote authentication systems.

7.1 Recommendations

Building upon our conclusions, we offer a set of recommendations that aim to shape the future of banknote authentication:

(i) **Technological Integration**: Promote the integration of advanced authentication technologies, such as blockchain, quantum-resistant encryption, and central bank digital currencies, into existing financial infrastructures.

(ii) **Data Privacy and Ethics:** Establish ethical guidelines and robust data protection practices for the responsible use of biometrics and personal data in banknote authentication.

(iii) **Interdisciplinary Insights**: Foster knowledge sharing and interdisciplinary research to leverage lessons from adjacent fields, such as cybersecurity and fintech.

(iv) **Sustainability Considerations:** Investigate the social and environmental sustainability of authentication technologies, including the materials and production processes used.

(v) **Tailored Solutions:** Develop banknote authentication solutions tailored to the unique challenges and requirements of small and developing economies.

(vi) **Enhanced User Experience**: Implement human-centered design principles to create user-friendly and trustworthy banknote authentication systems.

7.2 Future Enhancements

The future of banknote authentication is dynamic and multifaceted. As technology continues to advance and counterfeiting techniques become more sophisticated, the need for cutting-edge, adaptable solutions remains paramount. The integration of machine learning, blockchain, and biometric technologies is reshaping the security landscape, while the imperative to protect user data and privacy is increasingly critical.

APPENDIX A:TIME SCHEDULE

Month Task	*May-June*	*June-July*	July-Aug	Aug-Sep	Sep-oct	Oct-Nov	Nov-Dec
Proposal writing and Introduction							
Literature Review and Research methodology							
System Analysis							
System Design							
System implementation and testing							
Documentation							
Conclusion & Recommendation							

Figure 7: Time Schedule

APPENDIX B BUDGET

NO	ITEMS	QUANTITY	COST /UNIT	TOTAL COST(KSH)
1	Laptop	1	50,000	50,000
2	Flash drive 8gb	1	1500	1500
3	Kaspersky Antivirus	1	2700	2700
4	Operating system	1	2000	2000
5	Microsoft office suit	1	5000	5000
	Total			61,200

Figure 8: Budget

APPENDIX C:PROJECT CODE

Decision tree code:

```
class DecisionTree():
    def __init__(self, min_sample_split = 2, max_depth = 100, no_of_features = None):
        self.min_sample_split = min_sample_split
        self.max_depth = max_depth
        self.no_of_features = no_of_features
        self.root = None

    def fit(self, X, y):
        self.no_of_features = X.shape[1] if not self.no_of_features else min(self.no_of_features, X.shape[1])
        self.root = self._grow_tree(X, y)

    def _grow_tree(self, X, y, depth = 0):
        no_of_samples, no_of_features = X.shape
        no_of_labels = len(np.unique(y))

        #checking for the stopping criteria
        if(depth >= self.max_depth or no_of_labels == 1 or no_of_samples < self.min_sample_split):
            leaf_value = self._most_common_label(y)
            return Node(value = leaf_value)

        #Selecting an array of feature indices randomly from the given features
        feature_index = np.random.choice(no_of_features, self.no_of_features, replace = False)

        #greedy search for best splitting criteria
        best_feature, best_threshold = self._best_criteria(X, y, feature_index)
        left_idxs, right_idxs = self._split(X[:, best_feature], best_threshold)
        left = self._grow_tree(X[left_idxs, :], y[left_idxs], depth+1)
        right = self._grow_tree(X[right_idxs, :], y[right_idxs], depth+1)

        return Node(best_feature, best_threshold, left, right)

    def _best_criteria(self, X, y, feat_idxs):
        best_gain = -1
        split_index, split_threshold = None, None

        for feat_idx in feat_idxs:
            X_column = X[:, feat_idx]
            thresholds = np.unique(X_column)

            for threshold in thresholds:
                gain = self._information_gain(y, X_column, threshold)
```

```
            if gain > best_gain:
                best_gain = gain
                split_index = feat_idx
                split_threshold = threshold

    return split_index, split_threshold

def _information_gain(self, y, X_column, split_threshold):
    #calculate the parents entropy
    parent_entropy = entropy(y)

    #generate split
    left_idxs, right_idxs = self._split(X_column, split_threshold)

    if len(left_idxs) == 0 or len(right_idxs) == 0:
        return 0

    #calculate [weighted average]E(children)
    n = len(y)
    n_l, n_r = len(left_idxs), len(right_idxs)
    e_l, e_r = entropy(y[left_idxs]), entropy(y[right_idxs])
    child_entropy = (n_l/n)*e_l + (n_r/n)*e_r

    #return information gain = E(parent) - [weighted average]E(children)
    information_gain = parent_entropy - child_entropy

    return information_gain

def _split(self, X_column, split_threshold):
    #gives an array of all the indices in X_column that follow the splitting criteria
    left_idxs = np.argwhere(X_column <= split_threshold).flatten()
    right_idxs = np.argwhere(X_column > split_threshold).flatten()

    return left_idxs, right_idxs

def _most_common_label(self, y):
    counter = Counter(y)
    most_common = counter.most_common(1)[0][0]
    return most_common

def predict(self, X):
    return np.array([self._traverse_tree(x, self.root) for x in X])
```

```
  def _traverse_tree(self, x, node):
    if node.is_leaf_node():
      return node.value

    if x[node.feature] <= node.threshold:
      return self._traverse_tree(x, node.left)

    return self._traverse_tree(x, node.right)
```

KNN classifier

```
class KNN:
  def __init__(self, k = 3):
    self.k = k

  def fit(self, X, y):
    self.X_train = X
    self.y_train = y

  def predict(self, X):
    predicted_labels = [self._predict(x) for x in X]
    return np.array(predicted_labels)

  def _predict(self, x):
    distances = [euclidean_distance(x, x_train) for x_train in self.X_train]
    k_indices = np.argsort(distances)[:self.k]
    k_nearest_labels = [self.y_train[i] for i in k_indices]
    most_common = Counter(k_nearest_labels).most_common(1)
    return most_common[0][0]
```

FOR K=5

```
KNN_Classifier_5.fit(X_train_KNN, y_train_KNN)
y_predicted_KNN_5 = KNN_Classifier_5.predict(X_test_KNN)

accuracy_KNN_5 = accuracy(y_test_KNN, y_predicted_KNN_5)

accuracy_dict[3] = ["KNN (k = 5)", accuracy_KNN_5]
print("Accuracy of KNN for k = 5 is :", accuracy_KNN_5)
```

FOR K=3

```
KNN_Classifier_3 = KNN(k = 3)
KNN_Classifier_3.fit(X_train_KNN, y_train_KNN)
y_predicted_KNN_3 = KNN_Classifier_3.predict(X_test_KNN)

accuracy_KNN_3 = accuracy(y_test_KNN, y_predicted_KNN_3)
```

```
accuracy_dict[2] = ["KNN (k = 3)", accuracy_KNN_3]
print("Accuracy for k = 3 is :", accuracy_KNN_3)
```

RANDOM FOREST CLASSIFIER

```
class RandomForest():
    def __init__(self, no_of_trees=100, min_sample_split=2, max_depth=100,
no_of_features=None):
        self.no_of_trees = no_of_trees
        self.min_sample_split = min_sample_split
        self.max_depth = max_depth
        self.no_of_features = no_of_features
        self.trees = []

    def fit(self, X, y):
        self.trees = []
        for _ in range(self.no_of_trees):
            tree = DecisionTree(min_sample_split = self.min_sample_split, max_depth =
self.max_depth, no_of_features = self.no_of_features)
            X_sample, y_sample = bootstrap_sample(X, y)
            tree.fit(X_sample, y_sample)
            self.trees.append(tree)

    def predict(self, X):
        tree_predictions = np.array([tree.predict(X) for tree in self.trees])
        tree_predictions = np.swapaxes(tree_predictions, 0, 1)

        y_predictions = [most_common_label(tree_prediction) for tree_prediction in
tree_predictions]

        return np.array(y_predictions)
```

SVM CODE

```
class SVM:
    def __init__(self, learning_rate=0.001, lambda_param=0.01, n_iters=1000):
        self.lr = learning_rate
        self.lambda_param = lambda_param
        self.n_iters = n_iters
        self.w = None
        self.b = None

    def fit(self, X, y):
        n_samples, n_features = X.shape

        y_ = np.where(y <= 0, -1, 1)

        self.w = np.zeros(n_features)
        self.b = 0

        for _ in range(self.n_iters):
            for idx, x_i in enumerate(X):
                condition = y_[idx] * (np.dot(x_i, self.w) - self.b) >= 1
                if condition:
                    self.w -= self.lr * (2 * self.lambda_param * self.w)
                else:
                    self.w -= self.lr * (
                        2 * self.lambda_param * self.w - np.dot(x_i, y_[idx])
                    )
                    self.b -= self.lr * y_[idx]

    def predict(self, X):
        approx = np.dot(X, self.w) - self.b
        return np.sign(approx)

    def accuracy(y_true, y_predicted):
      accuracy = np.sum(y_true == y_predicted)/len(y_true)
      return accuracy clf = SVM()
clf.fit(X_train_svm, y_train_svm)
# predictions = clf.predict(X)

print(clf.w, clf.b)

def visualize_svm():
    def get_hyperplane_value(x, w, b, offset):
        return (-w[0] * x + b + offset) / w[1]
```

```
    fig = plt.figure()
    ax = fig.add_subplot(1, 1, 1)
    plt.scatter(X_svm[:, 0], X_svm[:, 1], marker="o", c=y_svm)

    x0_1 = np.amin(X_svm[:, 0])
    x0_2 = np.amax(X_svm[:, 0])

    x1_1 = get_hyperplane_value(x0_1, clf.w, clf.b, 0)
    x1_2 = get_hyperplane_value(x0_2, clf.w, clf.b, 0)

    x1_1_m = get_hyperplane_value(x0_1, clf.w, clf.b, -1)
    x1_2_m = get_hyperplane_value(x0_2, clf.w, clf.b, -1)

    x1_1_p = get_hyperplane_value(x0_1, clf.w, clf.b, 1)
    x1_2_p = get_hyperplane_value(x0_2, clf.w, clf.b, 1)

    ax.plot([x0_1, x0_2], [x1_1, x1_2], "y--")
    ax.plot([x0_1, x0_2], [x1_1_m, x1_2_m], "k")
    ax.plot([x0_1, x0_2], [x1_1_p, x1_2_p], "k")

    x1_min = np.amin(X_svm[:, 1])
    x1_max = np.amax(X_svm[:, 1])
    ax.set_ylim([x1_min - 3, x1_max + 3])

    plt.show()

visualize_svm()
```

TABULATED RESULTS

Sr. No.	Algorithm	Accuracy
1	Decision Tree	99.27%
2	KNN (k = 3)	96.0%
3	KNN (k = 5)	95.64%
4	Random forest	98.55%
5	Support Vector Machine	41.82%

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